ACADEMIC REGULATIONS & SYLLABUS

Faculty of Pharmacy
Master of Pharmacy Programme

(Pharmaceutical Chemistry)

(AS PER PCI SYLLABUS)

A.Y. 2022-2023

Ramanbhai Patel College of Pharmacy

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Vision of RPCP

To Become a Premier Pharma Institute by Creating World Class Pharmacists and Researchers.

Mission of RPCP

To Strive for the Excellence in Pharmaceutical Sciences through Quality Education and Research.

PROGRAM OUTCOMES

1. Pharmacy Knowledge: Possess knowledge and comprehension of the core and basic knowledge associated with the profession of pharmacy, including biomedical sciences; pharmaceutical sciences; behavioral, social, and administrative pharmacy sciences; and manufacturing practices.

2. Planning Abilities: Demonstrate effective planning abilities including time management, resource management, delegation skills and organizational skills. Develop and implement plans and organize work to meet deadlines.

3. Problem analysis: Utilize the principles of scientific enquiry, thinking analytically, clearly and critically, while solving problems and making decisions during daily practice. Find, analyze, evaluate and apply information systematically and shall make defensible decisions.

4. Modern tool usage: Learn, select, and apply appropriate methods and procedures, resources, and modern pharmacy-related computing tools with an understanding of the limitations.

5. Leadership skills: Understand and consider the human reaction to change, motivation issues, leadership and team-building when planning changes required for fulfilment of practice, professional and societal responsibilities. Assume participatory roles as responsible citizens or leadership roles when appropriate to facilitate improvement in health and wellbeing.

6. Professional Identity: Understand, analyze and communicate the value of their professional roles in society (e.g. health care professionals, promoters of health, educators, managers, employers, employees).

7. Pharmaceutical Ethics: Honour personal values and apply ethical principles in professional and social contexts. Demonstrate behavior that recognizes cultural and personal variability in values, communication and lifestyles. Use ethical frameworks; apply ethical principles while making decisions and take responsibility for the outcomes associated with the decisions.

8. Communication: Communicate effectively with the pharmacy community and with society at large, such as, being able to comprehend and write effective reports, make effective presentations and documentation, and give and receive clear instructions.

9. The Pharmacist and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety and legal issues and the consequent responsibilities relevant to the professional pharmacy practice.

10. Environment and sustainability: Understand the impact of the professional pharmacy solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

11. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. Self-assess and use feedback effectively from others to identify learning needs and to satisfy these needs on an ongoing basis.

FACULTY OF PHARMACY

ACADEMIC REGULATIONS
MASTER OF PHARMACY (M. Pharm.) PROGRAMME
Choice Based Credit System (CBCS)

1. Short Title and Commencement

These regulations shall be called as "The Revised Academic Regulations for the postgraduate programmes under the Faculty of Pharmacy". They shall come into effect from the Academic Year 2018-19. The regulations framed are subject to modifications from time to time by the respective regulatory bodies.

2. Minimum Qualification for Admission

2.1 Candidate shall have passed B. Pharm Degree examination of an Indian university established by law in India from an institution approved by Pharmacy Council of India and has scored not less than 55 % of the maximum marks (aggregate of 4 years of B.Pharm.)

2.2 Every student, selected for admission to post graduate pharmacy program in any PCI approved institution should have obtained registration with the State Pharmacy Council or should obtain the same within one month from the date of his/her admission, failing which the admission of the candidate shall be cancelled.

Note: It is mandatory to submit a migration certificate obtained from the respective university where the candidate had passed his/her qualifying degree (B.Pharm.).

3. Duration of the Programme

The course of study for M.Pharm shall extend over a period of four semesters (two academic years). The curricula and syllabi for the program shall be prescribed from time to time by Pharmacy Council of India, New Delhi.

4. Medium of Instruction and Examinations

Medium of instruction and examination shall be in English.

5. Working Days in a Semester

Each semester shall consist of not less than 100 working days. The odd semesters shall be conducted from the month of June/July to November/December and the even semesters shall be conducted from December/January to May/June in every calendar year.

6. Attendance and Progress

A candidate is required to put in at least 80% attendance in individual courses considering theory and practical separately. The candidate shall complete the prescribed course satisfactorily to be eligible to appear for the respective examinations.

7. Programme Credit Structure

As per the philosophy of Credit Based Semester System, certain quantum of academic work viz. theory classes, tutorial hours, practical classes, etc. are measured in terms of credits. On satisfactory completion of the courses, a candidate earns credits. The amount of credit associated with a course is dependent upon the number of hours of instruction per week in that course. Similarly, the credit associated with any of the other academic, co/extra-curricular activities is dependent upon the quantum of work expected to be put in for each of these activities per week.

7.1. Credit assignment

7.1.1. Theory and Laboratory courses

Courses are broadly classified as Theory and Practical. Theory courses consist of lecture (L) and Practical (P) courses consist of hours spent in the laboratory. Credits (C) for a course is dependent on the number of hours of instruction per week in that course, and is obtained by using a multiplier of one (1) for lecture and a multiplier of half (1/2) for practical (laboratory) hours. Thus, for example, a theory course having four lectures per week throughout the semester carries a credit of 4. Similarly, a practical having four laboratory hours per week throughout semester carries a credit of 2.

The contact hours of seminars, assignments and research work shall be treated as that of practical courses for the purpose of calculating credits. i.e., the contact hours shall be multiplied by 1/2. Similarly, the contact hours of journal club, research work presentations and discussions with the supervisor shall be considered as theory course and multiplied by 1.

7.2. Minimum credit requirements

The minimum credit points required for the award of M. Pharm. degree is 95. However based on the credit points earned by the students under the head of co-curricular activities, a student shall earn a maximum of 100 credit points. These credits are divided into Theory courses, Practical, Seminars, Assignments, Research work, Discussions with the supervisor, Journal club and Co-Curricular activities over the duration of four semesters. The credits are distributed semester-wise. Courses generally progress in sequence, building competencies and their positioning indicates certain academic maturity on the part of the learners. Learners are expected to follow the semester-wise schedule of courses given in the syllabus.

8. Academic work

A regular record of attendance both in Theory, Practical, Seminar, Assignment, Journal club, Discussion with the supervisor, Research work presentation and Dissertation shall be maintained by the department / teaching staff of respective courses**.**

9. Examinations/Assessments

The scheme for internal assessment and end semester examinations is given in Annexure II (Table – 1 to 5).

9.1. End Semester Examinations

The End Semester Examinations for each theory and practical course through semesters I to IV shall be conducted by the University for which Examinations shall be conducted by the subject experts at college level and the marks/grades shall be submitted to the university.

9.2. Internal Assessment: Continuous Mode

The marks allocated for Continuous mode of Internal Assessment shall be awarded as per the scheme given below.

Table- 1: Scheme for awarding internal assessment: Continuous mode

Theory	
Criteria	**Maximum Marks**
Attendance (Refer Table – 2)	8
Student – Teacher interaction	2
Total	10
Practical	
Attendance (Refer Table – 2)	10
Based on Practical Records, Regular viva voce, etc.	10
Total	20

Table- 2: Guidelines for the allotment of marks for attendance

Percentage of Attendance	**Theory**	**Practical**
95 – 100	8	10
90 – 94	6	7.5
85 – 89	4	5
80 – 84	2	2.5
Less than 80	0	0

9.2.1. Sessional Exams

Two Sessional exams shall be conducted for each theory / practical course as per the schedule fixed by the college. The scheme of question paper for theory and practical Sessional examinations will be as prescribed by the regulatory body. The average marks of two Sessional exams shall be computed for internal assessment as per the requirements given in Annexure II. Sessional exam shall be conducted for 30 marks for theory and shall be computed for 15 marks. Similarly Sessional exam for practical shall be conducted for 40 marks and shall be computed for 10 marks.

10. Promotion and Award of Grades

A student shall be declared PASS and eligible for getting grade in a course of M.Pharm. Programme, if he/she secures at least 50% marks in that particular course including internal assessment.

11. Carry Forward of Marks

In case a student fails to secure the minimum 50% in any Theory or Practical course as specified in 9 above, then he/she shall reappear for the end semester examination of that course. However his/her marks of the Internal Assessment shall be carried over and he/she shall be entitled for grade obtained by him/her on passing.

12. Improvement of Internal Assessment

A student shall have the opportunity to improve his/her performance only once in the Sessional exam component of the internal assessment. The re-conduct of the Sessional exam shall be completed before the commencement of next end semester theory examinations.

13. Re-examination of End Semester Examinations

Re-examination of end semester examination shall be conducted as per the schedule given in table 3. The exact dates of examinations shall be notified from time to time.

Table-3: Tentative schedule of end semester examinations

Semester	For Regular Candidates	For Failed Candidates
I and III	November / December	May / June
II and IV	May / June	November / December

14. Academic Progression

No student shall be admitted to any examination unless he/she fulfils the norms given in item no. 6 under the heading of attendance and progress. ATKT rules are applicable as follows:

A student shall be eligible to carry forward all the courses of I and II semesters till the III semester examinations. However, he/she shall not be eligible to attend the courses of IV semester until all the courses of I, II and III semesters are successfully completed.

A student shall be eligible to get his/her CGPA upon successful completion of the courses of I to IV semesters within the stipulated time period as per the norms.

Note: Grade AB should be considered as failed and treated as one head for deciding ATKT. Such rules are also applicable for those students who fail to register for examination(s) of any course in any semester.

15. Grading of Performances (Letter Grades and Grade Points Allocations)

Based on the performances, each student shall be awarded a final letter grade at the end of the semester for each course. The letter grades and their corresponding grade points are given in Table – 4.

Table – 4: Letter grades and grade points equivalent to Percentage of marks and performances

Percentage of Marks Obtained	Letter Grade	Grade Point	Performance
90.00 – 100	AA	10	Outstanding
80.00 – 89.99	AB	9	Excellent
70.00 – 79.99	BB	8	Good
60.00 – 69.99	BC	7	Fair
50.00 – 59.99	CC	6	Average
Less than 50	FF	0	Fail
Absent	Ab	0	Fail

A learner who remains absent for any end semester examination shall be assigned a letter grade of Ab. and a corresponding grade point of zero. He/she should reappear for the said evaluation/examination in due course.

16. Semester Grade Point Average (SGPA)

The performance of a student in a semester is indicated by a number called 'Semester Grade Point Average' (SGPA). The SGPA is the weighted average of the grade points obtained in all the courses by the student during the semester. For example, if a student takes five courses(Theory/Practical) in a semester with credits C1, C2, C3, C4 and C5 and the student's grade points in these courses are G1, G2, G3, G4 and G5, respectively, and then students' SGPA is equal to:

$$\text{SGPA}=\frac{C1G1+C2G2+C3G3+C4G4}{C1+C2+C3+C4}$$

The SGPA is calculated to two decimal points. It should be noted that, the SGPA for any semester shall take into consideration the FF and Ab. grade awarded in that semester. For example if a learner has a FF or Ab. grade in course 4, the SGPA shall then be computed as:

$$\text{SGPA}=\frac{C1G1+C2G2+C3G3+C4Zero}{C1+C2+C3+C4}$$

17. Cumulative Grade Point Average (CGPA)

The CGPA is calculated with the SGPA of all the VIII semesters to two decimal points and is indicated in final grade report card/final transcript showing the grades of all VIII semesters and their courses. The CGPA shall reflect the failed status in case of FF grade(s), till the course(s) is/are passed. When the course(s) is/are passed by obtaining a pass grade on subsequent examination(s) the CGPA shall only reflect the new grade and not the fail grades earned earlier. The CGPA is calculated as:

$$\textbf{CGPA}=\frac{\boldsymbol{C1S1+C2S2+C3S3+C4S4}}{\boldsymbol{C1+C2+C3+C4}}$$

where C1, C2, C3,…. is the total number of credits for semester I,II,III,…. and S1,S2, S3,….is the SGPA of semester I,II,III,…. .

No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.

18. Declaration of Class (Table-5)

The class shall be awarded on the basis of CGPA as follows:

First Class with Distinction	= CGPA of. 7.50 to 10.0
First Class	= CGPA of 6.0 to 7.49
Second Class	= CGPA of 5.0 to 5.99
Pass Class	=CGPA of 5.00

19. Project Work

All the students shall undertake a project under the supervision of a teacher in Semester III to IV and submit a report. 4 copies of the project report shall be submitted (typed & bound copy not less than 75 pages).

The internal and external examiner appointed by the University shall evaluate the project at the time of the Practical examinations of other semester(s). The projects shall be evaluated as per the criteria given below.

Evaluation of Dissertation Book			
Criteria	**Semester-III**	**Semester-IV**	
		Internal Evaluation (Marks)	**External Evaluation (Marks)**
Objective(s) of the work done	--	05	05
Methodology Adopted	--	25	25
Results and Discussions	--	15	15
Conclusions and Outcome	--	05	05
Total	--	50	50
Final Total		**100**	

Evaluation of Presentation				
	Semester-III		**Semester-IV**	
Criteria	**Internal Evaluation (Marks)**	**External Evaluation (Marks)**	**Internal Evaluation (Marks)**	**External Evaluation (Marks)**
Presentation of work	75	100	75	75
Communication skills	25	50	25	25
Question and answer skills	50	50	50	50
Total	150	200	150	150
Final Total	**350**		**300**	

20. Award of Ranks

Ranks and Medals shall be awarded on the basis of final CGPA. However, candidates who fail in one or more courses during the M. Pharm. program shall not be eligible for award of ranks. Moreover, the candidates should have completed the M. Pharm program in minimum prescribed number of years, (two years) for the award of Ranks.

21. Award of degree

Candidates who fulfil the requirements mentioned above shall be eligible for award of degree during the ensuing convocation.

22. Duration for completion of the program of study

The duration for the completion of the program shall be fixed as double the actual duration of the program and the students have to pass within the said period, otherwise they have to get fresh Registration.

23. Extra Credit:

An extra credit is to be offered to a student for achievements in co-curricular and extra–curricular activities. This credit shall not be counted while considering the minimum credits for completing the program. The activities and appropriate weight (points) to be allocated to award an extra credit are broadly classified as per the table below:

Sr. no.	Name of the Activity	Maximum Credit Points Eligible / Activity
1	Participation in National Level Seminar/Conference/Workshop/Symposium/ Training Programs (related to the specialization of the student)	01
2	Participation in international Level Seminar/Conference/Workshop/Symposium/ Training Programs (related to the specialization of the student)	02
3	Academic Award/Research Award from State Level/National Agencies	01
4	Academic Award/Research Award from International Agencies	02
5	Research / Review Publication in National Journals (Indexed in Scopus / Web of Science)	01
6	Research / Review Publication in International Journals (Indexed in Scopus / Web of Science)	02

Note: International Conference: Held Outside India

International Journal: The Editorial Board Outside India

*The credit points assigned for extracurricular and or co-curricular activities shall be given by the Principals of the colleges and the same shall be submitted to the University. The criteria to acquire this credit point shall be defined by the colleges from time to time.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

MASTER OF PHARMACY (PHARMACEUTICAL CHEMISTRY) PROGRAMME

Schemes for internal assessments and end semester examinations

SEMESTER-I

SCHEME OF TEACHING

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
PHCCC009	Modern Pharmaceutical Analytical Techniques	4	4	4
PHPCM001	Advanced Organic Chemistry-I	4	4	4
PHPCM002	Advanced Medicinal Chemistry	4	4	4
PHPCM003	Chemistry of Natural Products	4	4	4
PHPCM004	Pharmaceutical Chemistry Practical – I	12	6	12
PHPCM005	Seminar/Assignment-I	2	1	2
---	DHSS Elective-I*	2	2	2
---	University Elective-I**	2	2	2
	Total	**34**	**27**	**34**

SCHEME OF EVALUATION

Course Code	Course	Internal Assessment				End Semester Exams		Total Marks
		Continuous Mode	Sessional Exams		Total	Marks	Duration	
			Marks	Duration				
PHCCC009	Modern Pharmaceutical Analytical Techniques	10	15	1Hr	**25**	**75**	3Hrs	**100**
PHPCM001	Advanced Organic Chemistry-I	10	15	1Hr	**25**	**75**	3Hrs	**100**
PHPCM002	Advanced Medicinal Chemistry	10	15	1Hr	**25**	**75**	3Hrs	**100**
PHPCM003	Chemistry of Natural Products	10	15	1Hr	**25**	**75**	3Hrs	**100**
PHPCM004	Pharmaceutical Chemistry Practical - I	20	30	6Hrs	**50**	**100**	6Hrs	**150**
PHPCM005	Seminar/ Assignment-I	-	-	-	**100**	**-**	-	**100**
---	DHSS elective-I*	-	-	-	**30**	**70**	-	**100**
---	University Elective-I**	-	-	-	**30**	**70**	-	**100**
Total								**850**

*DHSS elective courses: SCHEME OF TEACHING

Semester	Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
I	HS704 A/B/C/D/E/F/G/H	Academic Speaking	02	02	02

*DHSS elective courses: SCHEME OF EVALUATION

Course Code	Course Name	Evaluation Scheme				
		Theory		Practical		Total
		Internal	External	Internal	External	
HS704 A/B/C/D/E/F/G/H	Academic Speaking	-	-	30	70	100

**University elective courses: SCHEME OF TEACHING - Semester-I

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
MA771.01	Reliability and Risk Analysis-I	2	2	2
EE781.01	Optimization Techniques	2	2	2
CE772.01	Research Methodology	2	2	2
CA730	Internet and Web Designing	2	2	2
PT795.01	Heath and Physical activity	2	2	2
NR755	First aid and Life support	2	2	2
RD701.01	Introduction to analytical Techniques	2	2	2
RD702.01	Introduction to Nanoscience and Technology	2	2	2
MB650	Creative Leadership	2	2	2
PH891	Community Pharmacy ownership	2	2	2
ME781.01	Occupational Health and Safety	2	2	2
PD261	Astrophysics, Space and Cosmos-1(ASC-1)	2	2	2

**University elective courses: SCHEME OF EVALUATION- Semester-I

Course Code	Course Name	Evaluation Scheme				
		Theory		Practical		Total
		Internal	External	Internal	External	
MA771.01	Reliability and Risk Analysis-I	-	-	30	70	100
EE781.01	Optimization Techniques	-	-	30	70	100
CE772.01	Research Methodology	-	-	30	70	100
CA730	Internet and Web Designing	-	-	30	70	100
PT795.01	Heath and Physical activity	-	-	30	70	100
NR755	First aid and Life support	-	-	30	70	100
RD701.01	Introduction to analytical Techniques	-	-	30	70	100
RD702.01	Introduction to Nanoscience and Technology	-	-	30	70	100
MB650	Creative Leadership	-	-	30	70	100
PH891	Community Pharmacy ownership	-	-	30	70	100
ME781.01	Occupational Health and Safety	-	-	30	70	100
PD261	Astrophysics, Space and Cosmos-1(ASC-1)	-	-	30	70	100

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
MASTER OF PHARMACY (PHARMACEUTICAL CHEMISTRY) PROGRAMME
Schemes for internal assessments and end semester examinations

SEMESTER-II
SCHEME OF TEACHING

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
PHPCM006	Advanced Spectral Analysis	4	4	4
PHPCM007	Advanced Organic Chemistry-II	4	4	4
PHPCM008	Computer Aided Drug Design	4	4	4
PHPCM009	Pharmaceutical Process Chemistry	4	4	4
PHPCM010	Pharmaceutical Chemistry Practical-II	12	6	12
PHPCM011	Seminar/Assignment-II	2	1	2
---	DHSS elective-II*	2	2	2
---	University Elective-II**	2	2	2
	Total	**34**	**27**	**34**

SCHEME OF EVALUATION

Course Code	Course	Internal Assessment				End Semester Exams		Total Marks
		Continuous Mode	Sessional Exams		Total	Marks	Duration	
			Marks	Duration				
PHPCM006	Advanced Spectral Analysis	10	15	1Hr	25	75	3Hrs	100
PHPCM007	Advanced Organic Chemistry-II	10	15	1Hr	25	75	3Hrs	100
PHPCM008	Computer Aided Drug Design	10	15	1Hr	25	75	3Hrs	100
PHPCM009	Pharmaceutical Process Chemistry	10	15	1Hr	25	75	3Hrs	100
PHPCM010	Pharmaceutical Chemistry Practical-II	20	30	6Hrs	50	100	6Hrs	150
PHPCM011	Seminar/Assignment-II	-	-	-	100	-	-	100
---	DHSS elective-II*	-	-	-	30	70	-	100
---	University Elective-II**	-	-	-	30	70	-	100
Total								**850**

*DHSS elective courses: SCHEME OF TEACHING

Semester	Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
II	HS705 (B)	Academic Writing	02	02	02

*DHSS elective courses: SCHEME OF EVALUATION

Course Code	Course Name	Evaluation Scheme				
		Theory		Practical		Total
		Internal	External	Internal	External	
HS705 (B)	Academic Writing	-	-	30	70	100

**University elective courses: SCHEME OF TEACHING - Semester-II

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
EE782.01	Energy Auditing and Management	2	2	2
CE771.01	Project Management	2	2	2
IT771.01	Cyber Security & Laws	2	2	2
CA842	Mobile Application Development	2	2	2
PT796.01	Fitness and Nutrition	2	2	2
NR 752.01	Epidemiology & Community Heath	2	2	2
OC733.01	Introduction to Polymer Science	2	2	2
MB651	Software based Statistical Analysis	2	2	2
MA772.01	Design of Experiments	2	2	2
PH892	Intellectual Property Rights	2	2	2
PD262	Astrophysics, Space and Cosmos-2 (ASC-2)	2	2	2

**University elective courses: SCHEME OF EVALUATION- Semester-II

Course Code	Course Name	Evaluation Scheme				
		Theory		Practical		Total
		Internal	External	Internal	External	
EE782.01	Energy Auditing and Management	-	-	30	70	100
CE771.01	Project Management	-	-	30	70	100
IT771.01	Cyber Security & Laws	-	-	30	70	100
CA842	Mobile Application Development	-	-	30	70	100
PT796.01	Fitness and Nutrition	-	-	30	70	100
NR 752.01	Epidemiology & Community Heath	-	-	30	70	100
OC733.01	Introduction to Polymer Science	-	-	30	70	100
MB651	Software based Statistical Analysis	-	-	30	70	100
MA772.01	Design of Experiments	-	-	30	70	100
PH892	Intellectual Property Rights	-	-	30	70	100
PD262	Astrophysics, Space and Cosmos-2 (ASC-2)	-	-	30	70	100

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
MASTER OF PHARMACY (PHARMACEUTICAL CHEMISTRY) PROGRAMME
Schemes for internal assessments and end semester examinations

SEMESTER-III

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
PHCCC010	Research Methodology and Biostatistics*	4	4	4
PHCCC005	Journal Club-I	01	01	01
PHPCM012	Discussion / Presentation (Proposal Presentation)	02	02	02
PHPCM013	Research Work-I	28	14	28
	Total	**35**	**21**	**35**

***Non University Examination**

SCHEME OF EVALUATION

Course Code	Course	Internal Assessment				End Semester Exams		Total Marks
		Continuous Mode	Sessional Exams		Total	Marks	Duration	
			Marks	Duration				
PHCCC010	Research Methodology and Biostatistics	10	**15**	**1**	**25**	**75**	**3**	**100**
PHPCM012	Discussion / Presentation (Proposal Presentation)	-	-	-	50	-	-	50
PHCCC005	Journal Club-I	-	-	-	25	-	-	25
PHPCM013	Research Work-I				-	350	1hr	350
Total								**525**

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
MASTER OF PHARMACY (PHARMACEUTICAL CHEMISTRY) PROGRAMME
Schemes for internal assessments and end semester examinations

SEMESTER -IV

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
PHCCC006	Journal Club-II	1	1	1
PHPCM015	Research Work-II	31	16	31
PHPCM014	Discussion/Presentation	03	03	03
	Total	**35**	**20**	**35**

SCHEME OF EVALUATION

Course Code	Course	Internal Assessment				End Semester Exams		Total Marks
		Continuous Mode	Sessional Exams		Total	Marks	Duration	
			Marks	Duration				
PHPCM014	Discussion/ Presentation	-	-	-	75	-	-	75
PHCCC006	Journal Club-II	-	-	-	25	-	-	25
PHPCM015	Research Work-II				-	400	1hr	400
Total								**500**

Semester	Credit Point
I	27
II	27
III	21
IV	20
Co-curricular Activity (Attending Conference, Scientific Presentations, Publication, Industry Training and other scholarly Activities)	Maximum 5
Total Credit Point	**Minimum- 95** **Maximum-100***

***Credit Point for co-curricular activity**

Syllabi
Semester I

PHCCC009: MODERN PHARMACEUTICAL ANALYTICAL TECHNIQUES (Theory)

Total hours: 60 Hr

Outline of the course:

<table>
<tr><th>Sr. No.</th><th>Title of the unit</th><th>Minimum number of hours</th></tr>
<tr><td rowspan="4">1</td><td>UV-Visible spectroscopy</td><td rowspan="4">10</td></tr>
<tr><td>IR spectroscopy</td></tr>
<tr><td>Spectrofluorimetry</td></tr>
<tr><td>Flame emission spectroscopy and Atomic absorption spectroscopy</td></tr>
<tr><td>2</td><td>NMR Spectroscopy</td><td>10</td></tr>
<tr><td>3</td><td>Mass Spectroscopy</td><td>10</td></tr>
<tr><td>4</td><td>Chromatography</td><td>10</td></tr>
<tr><td rowspan="2">5</td><td>Electrophoresis</td><td rowspan="2">10</td></tr>
<tr><td>X-Ray Crystallography</td></tr>
<tr><td rowspan="2">6</td><td>Potentiometry</td><td rowspan="2">10</td></tr>
<tr><td>Thermal Techniques</td></tr>
</table>

Detailed Syllabus:

<table>
<tr><td>1.</td><td>UV-Visible spectroscopy</td><td rowspan="4">10 Hours</td><td rowspan="4">16.67%</td></tr>
<tr><td rowspan="3"></td><td>• UV-Visible spectroscopy: Introduction, Theory, Laws, Instrumentation associated with UV-Visible spectroscopy, Choice of solvents and solvent effect and Applications of UV Visible spectroscopy.</td></tr>
<tr><td>IR spectroscopy</td></tr>
<tr><td>• Theory, Modes of Molecular vibrations, Sample handling, Instrumentation of Dispersive and Fourier - Transform IR Spectrometer, Factors affecting vibrational frequencies and Applications of IR spectroscopy</td></tr>
</table>

	Spectrofluorimetry		
	Theory of Fluorescence, Factors affecting fluorescence, Quenchers, Instrumentation and Applications of fluorescence spectrophotometer.		
	Flame emission spectroscopy and Atomic absorption spectroscopy		
	• Principle, Instrumentation, Interferences and Applications.		
2.	**NMR Spectroscopy**	**10 Hours**	**16.67%**
	• Quantum numbers and their role in NMR, Principle, Instrumentation, Solvent requirement in NMR, Relaxation process, NMR signals in various compounds, Chemical shift, Factors influencing chemical shift, Spin-Spin coupling, Coupling constant, Nuclear magnetic double resonance, Brief outline of principles of FT-NMR and 13C NMR. Applications of NMR spectroscopy.		
3.	**Mass Spectroscopy**	**10 Hours**	**16.67%**
	• Mass Spectrometry: Principle, Theory, Instrumentation of Mass Spectrometry, Different types of ionization like electron impact, chemical, field, FAB and MALDI, APCI, ESI, APPI Analysers of Quadrupole and Time of Flight, Mass fragmentation and its rules, Meta stable ions, Isotopic peaks and Applications of Mass spectrometry.		
4.	**Chromatography**	**10 Hours**	**16.67%**
	• Principle, apparatus, instrumentation, chromatographic parameters, factors affecting resolution and applications of the following: a) Thin Layer chromatography b) High Performance Thin Layer Chromatography c) Ion exchange chromatography d) Column chromatography e) Gas chromatography f) High Performance Liquid chromatography g) Ultra High Performance Liquid chromatography h) Affinity chromatography i) Gel Chromatography		

5.	**Electrophoresis**	**10 Hours**	**16.67%**
	• Principle, Instrumentation, Working conditions, factors affecting separation and applications of the following: a) Paper electrophoresis b) Gel electrophoresis c) Capillary electrophoresis d) Zone electrophoresis e) Moving boundary electrophoresis f) Isoelectric focusing		
	X-ray Crystallography		
	• Production of X-rays, Different X-ray diffraction methods, Bragg's law, Rotating crystal technique, X-ray powder technique, Types of crystals and applications of X-ray diffraction.		
6.	**Potentiometry**	**10 Hours**	**16.67%**
	• Principle, working, Ion selective Electrodes and Application of potentiometry		
	Thermal Techniques		
	• Differential Scanning Calorimetry (DSC): Principle, thermal transitions and Instrumentation (Heat flux and power-compensation and designs), Modulated DSC, Hyper DSC, experimental parameters (sample preparation, experimental conditions, calibration, heating and cooling rates, resolution, source of errors) and their influence, advantage and disadvantages, pharmaceutical applications. • Differential Thermal Analysis (DTA): Principle, instrumentation and advantage and disadvantages, pharmaceutical applications, derivative differential thermal analysis (DDTA). TGA: Principle, instrumentation, factors affecting results, advantage and disadvantages, pharmaceutical applications.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	describe theory and principle of various spectroscopic and chromatographic separation techniques.
CO2	describe instrumentation and application of various spectroscopic and chromatographic separation techniques with justification.
CO3	summarize approaches to be adopted for quantitative & qualitative analysis of drugs in single and combine dosage forms.
CO4	Describe use of thermal methods and potentiometry in analysis of drugs/formulations
CO5	interpret the spectra and propose the structure of organic compounds.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	-	-	-	3
CO2	3	3	3	-	3	-	-	3	-	3	3
CO3	3	3	3	3	-	3	-	3	-	3	3
CO4	3	3	3	3	-	-	-	3	-	-	3
CO5	3	-	3	3	-	-	-	3	-	-	3

❖ **References:**

1. Spectrometric Identification of Organic compounds - Robert M Silverstein, Sixth edition, John Wiley & Sons, 2004.
2. Principles of Instrumental Analysis - Doglas A Skoog, F. James Holler, Timothy A. Nieman, 5th edition, Eastern press, Bangalore, 1998
3. Instrumental methods of analysis – Willards, 7th edition, CBS publishers.
4. Practical Pharmaceutical Chemistry – Beckett and Stenlake, Vol II, 4th edition, CBS Publishers, New Delhi, 1997.
5. Organic Spectroscopy - William Kemp, 3rd edition, ELBS, 1991.
6. Quantitative Analysis of Drugs in Pharmaceutical formulation - P D Sethi, 3rd Edition, CBS Publishers, New Delhi, 1997.

7. Pharmaceutical Analysis- Modern methods – Part B - J W Munson, Volume 11, Marcel Dekker Series. 8. The Analysis of Drugs in Biological Fluids, Joseph Chamberlain, CRC Press

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM001: ADVANCED ORGANIC CHEMISTRY-I (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Basic Aspects of Organic Chemistry	12
2	Study of mechanism and synthetic applications of named Reactions	12
3	Synthetic Reagents & Applications and Protection-Deprotection of functional group	12
4	Heterocyclic Chemistry	12
5	Synthon approach and retrosynthesis applications	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Basic Aspects of Organic Chemistry**	**12**	**20%**
	a. Organic intermediates: Carbocations, carbanions, free radicals, carbenes and nitrenes. Their method of formation, stability and synthetic applications. b. Types of reaction mechanisms and methods of determining them c. Detailed knowledge regarding the reactions, mechanisms and their relative reactivity and orientations. d. Nucleophilic uni- and bimolecular reactions (SN_1 and SN_2) Elimination reactions (E1 & E2; Hoffman & Saytzeff's rule) Rearrangement reaction		
2.	**Study of mechanism and synthetic applications of following named reactions**	**12**	**20%**
	Ugi reaction, Brook rearrangement, Ullmann coupling reactions, Dieckmann Reaction, Doebner-Miller Reaction, Sandmeyer		

	Reaction, Mitsunobu reaction, Mannich reaction, Vilsmeyer-Haack Reaction, Sharpless asymmetric epoxidation, Baeyer-Villiger oxidation, Shapiro & Suzuki reaction, Ozonolysis and Michael addition reaction		
3.	**Synthetic Reagents & Applications**	**12**	**20%**
	Aluminiumisopropoxide, N-bromosuccinamide, diazomethane, dicyclohexylcarbodimide, Wilkinson reagent, Witting reagent. Osmium tetroxide, titanium chloride, diazopropane, diethyl azodicarboxylate, Triphenylphosphine, Benzotriazol-1-yloxy) tris (dimethylamino) phosphonium hexafluoro-phosphate (BOP). **3.1 Protecting groups** Role of protection in organic synthesis Protection for the hydroxyl group, including 1,2-and1,3-diols: ethers, esters, carbonates, cyclic acetals & ketals Protection for the Carbonyl Group: Acetals and Ketals Protection for the Carboxyl Group: amides and hydrazides, esters Protection for the Amino Group and Amino acids: carbamates and amides		
4.	**Heterocyclic Chemistry**	**12**	**20%**
	4.1 Organic Name reactions with their respective mechanism and application involved in synthesis of drugs containing five, six membered and fused heterocyclic such as Debus-Radziszewski imidazole synthesis, Knorr Pyrazole Synthesis Pinner Pyrimidine Synthesis, Combes Quinoline Synthesis, Bernthsen Acridine Synthesis, Smiles rearrangement and Traube purine synthesis. **4.2** Synthesis of few representative drugs containing these heterocyclic nucleus such as Ketoconazole, Metronidazole, Miconazole, celecoxib, antipyrin, Metamizole sodium, Terconazole, Alprazolam, Triamterene, Sulfamerazine, Trimethoprim, Hydroxychloroquine, Quinine, Chloroquine,		

	Quinacrine, Amsacrine, Prochlorpherazine, Promazine, Chlorpromazine, Theophylline, Mercaptopurine and Thioguanine.		
5.	**Synthon approach and retrosynthesis applications**	**12**	**20%**
	a. Basic principles, terminologies and advantages of retrosynthesis; guidelines for dissection of molecules. b. Functional group interconversion and addition (FGI and FGA) c. C-X disconnections; C-C disconnections – alcohols and carbonyl compounds; 1,2-, 1,3-,1,4-, 1,5-, 1,6-difunctionalized compounds d. Strategies for synthesis of three, four, five and six-membered ring.		

Course Outcome (COs):

CO1	describe the mechanism & applications of various named reactions
CO2	understand the chemistry of heterocyclic compounds
CO3	narrate concept of disconnection to develop synthetic routes for small target molecule.
CO4	summarize various catalysts used in organic reactions

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	3	3

Recommended Study Material:

❖ **Text book:**

1. "Advanced Organic chemistry, Reaction, Mechanisms and Structure", J March, John Wiley and Sons, New York.
2. "Mechanism and Structure in Organic Chemistry", ES Gould, Hold Rinchart and Winston, New York.

3. “Organic Chemistry” Clayden, Greeves, Warren and Woihers., Oxford University Press 2001.
4. “Organic Chemistry” Vol I and II. I.L. Finar. ELBS, Pearson Education Lts, Dorling Kindersley 9 India) Pvt. Ltd.
5. A guide to mechanisms in Organic Chemistry, Peter Skyes (Orient Longman, New Delhi).
6. Reactive Intermediates in Organic Chemistry, Tandom and Gowel, Oxford & IBH Publishers.
7. Combinational Chemistry – Synthesis and applications – Stephen R Wilson & Anthony W Czarnik, Wiley – Blackwell.
8. Carey, Organic Chemistry, 5th Edition (Viva Books Pvt. Ltd.)
9. Organic Synthesis - The Disconnection Approach, S. Warren, Wily India
10. Principles of Organic Synthesis, ROC Norman and JM Coxan, Nelson Thorns.
11. Organic Synthesis - Special Techniques. VK Ahluwalia and R Agarwal, Narosa Publishers.

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM002: ADVANCED MEDICINAL CHEMISTRY (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Drug Discovery and Drug Target	12
2	Prodrug Design and Analog design	12
3	Advanced Medicinal Chemistry of Selective Class of Drug	12
4	Rational Design of Enzyme Inhibitors	12
5	Peptidomimetics	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Drug Discovery and Drug Target**	**12**	**20%**
	Drug discovery: Stages of drug discovery, lead discovery; identification, validation and diversity of drug targets. Biological drug targets: Receptors, types, binding and activation, theories of drug receptor interaction, drug receptor interactions, agonists vs antagonists, artificial enzymes.		
2.	**Prodrug Design and Analog design**	**12**	**20%**
	2.1 Prodrug design: Basic concept, Carrier linked prodrugs/ Bioprecursors, Prodrugs of functional group, Prodrugs to improve patient acceptability, Drug solubility, Drug absorption and distribution, site specific drug delivery and sustained drug action. Rationale of prodrug design and practical consideration of prodrug design. **2.2** Combating drug resistance: Causes for drug resistance,		

	strategies to combat drug resistance in antibiotics and anticancer therapy, Genetic principles of drug resistance. C **2.3** Analog Design: Introduction, Classical & Non classical, Bioisosteric replacement strategies, rigid analogs, alteration of chain branching, changes in ring size, ring position isomers, design of stereo isomers and geometric isomers, fragments of a lead molecule, variation in inter atomic distance.		
3.	**Advanced Medicinal Chemistry of Selective Class of Drug**	**12**	**20%**
	Systematic study, SAR, Mechanism of action and synthesis of new generation molecules of following class of drugs: **3.1** Anti-hypertensive drugs, Psychoactive drugs, Anticonvulsant drugs, H1 & H2 receptor antagonist, COX1 & COX2 inhibitors, Adrenergic & Cholinergic agents, Antineoplastic and Antiviral agents. **3.2** Stereochemistry and Drug action: Realization that stereo selectivity is a pre-requisite for evolution. Role of chirality in selective and specific therapeutic agents. Case studies, Enantio-selectivity in drug adsorption, metabolism, distribution and elimination.		
4.	**Rational Design of Enzyme Inhibitors**	**12**	**20%**
	Enzyme kinetics & Principles of Enzyme inhibitors, Enzyme inhibitors in medicine, Enzyme inhibitors in basic research, rational design of non-covalently and covalently binding enzyme inhibitors.		
5.	**Peptidomimetics**	**12**	**20%**
	Therapeutic values of Peptidomimetics, design of peptidomimetics by manipulation of the amino acids, modification of the peptide backbone, incorporating conformational constraints locally or globally. Chemistry of prostaglandins, leukotrienes and thromboxones.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	Describe different stages and techniques of Drug Discovery and Development
CO2	Summarize various strategies to design and develop new drug like molecules for biological targets
CO3	Understand role of medicinal chemistry in drug research
CO4	Narrate designing strategy and therapeutical value of peptidomimetics.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	3	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	3	-	-	-	-	-	-	3

Recommended Study Material:

- **Text book:**

  1. Medicinal Chemistry by Burger, Vol I –VI. 2.

  2. Wilson and Gisvold's Text book of Organic Medicinal and Pharmaceutical Chemistry, 12th Edition, Lippincott Williams & Wilkins, Woltess Kluwer (India) Pvt.Ltd, New Delhi.

  3. Comprehensive Medicinal Chemistry – Corwin and Hansch.

  4. Computational and structural approaches to drug design edited by Robert M Stroud and Janet. F Moore 81.

  5. Introduction to Quantitative Drug Design by Y.C. Martin.

  6. Principles of Medicinal Chemistry by William Foye, 7th Edition, Ippincott Williams & Wilkins, Woltess Kluwer (India) Pvt.Ltd, New Delhi.

  7. Drug Design Volumes by Arienes, Academic Press, Elsevier Publishers, Noida, Uttar Pradesh.

  8. Principles of Drug Design by Smith.

9. The Organic Chemistry of the Drug Design and Drug action by Richard B. Silverman, II Edition, Elsevier Publishers, New Delhi.
10. An Introduction to Medicinal Chemistry, Graham L.Patrick, III Edition, Oxford University Press, USA.
11. Biopharmaceutics and pharmacokinetics, D.M. Brahmankar, Sunil B. Jaiswal II Edition, 2014, Vallabh Prakashan, New Delhi.
12. Peptidomimetics in Organic and Medicinal Chemistry by Antonio Guarna and Andrea Trabocchi, First edition, Wiley publishers.

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM003: CHEMISTRY OF NATURAL PRODUCTS (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Study of Natural products as leads for new pharmaceuticals for the following class of drugs	12
2	Alkaloids, Flavonoids and Steroids	12
3	Terpenoids and Vitamins	12
4	Recombinant DNA technology and examples of crude drug use in Diabetic therapy	12
5	Structural elucidation of natural compounds	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Study of Natural products as leads for new pharmaceuticals for the following class of drugs**	**12**	**20%**
	a) Drugs Affecting the Central Nervous System: Morphine Alkaloids b) Anticancer Drugs: Paclitaxel and Docetaxel, Etoposide, and Teniposide c) Cardiovascular Drugs: Lovastatin, Teprotide and Dicoumarol d) Neuromuscular Blocking Drugs: Curare alkaloids e) Anti-malarial drugs and Analogues f) Chemistry of macrolide antibiotics (Erythromycin, Azithromycin, Roxithromycin, and Clarithromycin) and β - Lactam antibiotics (Cephalosporins and Carbapenem)		

2.	**Alkaloids, Flavonoids and Steroids**	**12**	**20%**
	General introduction, classification, isolation, purification, molecular modification and biological activity of alkaloids, general methods of structural determination of alkaloids, structural elucidation and stereochemistry of ephedrine, morphine, ergot, emetine and reserpine. Introduction, isolation and purification of flavonoids, General methods of structural determination of flavonoids; Structural elucidation of quercetin. General introduction, chemistry of sterols, sapogenin and cardiac glycosides. Stereochemistry and nomenclature of steroids, chemistry of contraceptive agents male & female sex hormones (Testosterone, Estradiol, Progesterone), adrenocorticoids (Cortisone), contraceptive agents and steroids (Vit – D).		
3.	**Terpenoids and Vitamins**	**12**	**20%**
	Classification, isolation, isoprene rule and general methods of structural elucidation of Terpenoids; Structural elucidation of drugs belonging to mono- (citral, menthol, camphor), di-(retinol, Phytol, taxol) and tri- terpenoids (Squalene, Ginsenoside) carotinoids (β carotene). Chemistry and Physiological significance of Vitamin A, B_1, B_2, B_{12}, C, E, Folic acid and Niacin.		
4.	**Recombinant DNA technology and examples of crude drug use in Diabetic therapy**	**12**	**20%**
	4.1 rDNA technology, hybridoma technology, New pharmaceuticals derived from biotechnology; Oligonucleotide therapy. Gene therapy: Introduction, Clinical application and recent advances in gene therapy, principles of RNA & DNA estimation **4.2** Active constituent of certain crude drugs used in Indigenous system Diabetic therapy – *Gymnema sylvestre*, *Salacia reticulate*,		

	Pterocarpus marsupiam, Swertia chirata, Trigonella foenum graccum; Liver dysfunction – *Phyllanthus niruri;* Antitumor – *Curcuma longa Linn*.		
5.	**Structural elucidation of natural compounds**	**12**	**20%**
	Structural characterization of natural compounds using IR, ^{1}HNMR, ^{13}CNMR and MS Spectroscopy of specific drugs e.g., Penicillin, Morphine, Camphor, Vit-D, Quercetin and Digitalis glycosides.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	summarize different types of natural compounds and their chemistry and medicinal importance.
CO2	describe the importance of natural compounds as lead molecules for new drug discovery.
CO3	comprehend the concept of rDNA technology tool for new drug discovery.
CO4	narrate general methods of structural elucidation of compounds of natural origin.
CO5	understand isolation, purification and characterization of simple chemical constituents from natural source.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	3	-	-	-	-	-	-	-	3
CO5	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

- **Text book:**

1. Modern Methods of Plant Analysis, Peech and M.V.Tracey, Springer – Verlag, Berlin, Heidelberg.
2. Phytochemistry Vol. I and II by Miller, Jan Nostrant Rein Hld.
3. Recent advances in Phytochemistry Vol. I to IV – Scikel Runeckles, Springer Science & Business Media.
4. Chemistry of natural products Vol I onwards IWPAC.
5. Natural Product Chemistry Nakanishi Gggolo, University Science Books, California.
6. Natural Product Chemistry "A laboratory guide" – Rapheal Khan.
7. The Alkaloid Chemistry and Physiology by RHF Manske, Academic Press.
8. Introduction to molecular Phytochemistry – CHJ Wells, Chapmannstall.
9. Organic Chemistry of Natural Products Vol I and II by Gurdeep and Chatwall, Himalaya Publishing House.
10. Organic Chemistry of Natural Products Vol I and II by O.P. Agarwal, Krishan Prakashan.
11. Organic Chemistry Vol I and II by I.L. Finar, Pearson education.
12. Elements of Biotechnology by P.K. Gupta, Rastogi Publishers.
13. Pharmaceutical Biotechnology by S.P.Vyas and V.K.Dixit, CBS Publishers.
14. Biotechnology by Purohit and Mathur, Agro-Bios, 13th edition.
15. Phytochemical methods of Harborne, Springer, Netherlands.
16. Burger's Medicinal Chemistry.

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM004: PHARMACEUTICAL CHEMISTRY PRACTICAL - I
(Practical)

Total hours: 180 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Analysis of Pharmacopoeial compounds and their formulations by UV Vis spectrophotometer, RNA and DNA estimation.
2	Simultaneous estimation of multi component containing formulations by UV spectrophotometry.
3	Experiments based on Column chromatography.
4	Experiments based on HPLC.
5	Experiments based on Gas Chromatography.
6	Estimation of riboflavin/quinine sulphate by fluorimetry.
7	Estimation of sodium/potassium by flame photometry.
8	Purification of organic solvents, column chromatography.
9	To perform the following reactions of synthetic importance Claisen-schimidt reaction Benzyllic acid rearrangement. Beckmann rearrangement. Hoffmann rearrangement Mannich reaction
10	Synthesis of medicinally important compounds involving more than one step along with purification and Characterization using TLC, melting point and IR spectroscopy.
11	Estimation of elements and functional groups in organic natural compounds
12	Isolation, characterization like melting point, mixed melting point, molecular weight determination, functional group analysis, co-chromatographic technique for identification of isolated compounds and interpretation of UV and IR data.
13	Some typical degradation reactions to be carried on selected plant constituents.

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	perform and optimize medicinally important name reaction.
CO2	purify and characterize the medicinally important synthesized compounds
CO3	perform the analysis of drug/formulation using analytical instruments like HPTLC, HPLC, UV/vis spectrometer etc.
CO4	analyze the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	-	3	-	-	-	-	-	3	-
CO2	3	3	-	-	-	-	-	-	-	3	3
CO3	3	3	-	3	-	-	-	-	-	3	3
CO4	-	3	3	-	3	-	-	-	-	-	-

Recommended Study Material:

- ❖ **Text book:**

  1. "Advanced Organic chemistry, Reaction, Mechanisms and Structure", J March, John Wiley and Sons, New York.
  2. "Mechanism and Structure in Organic Chemistry", ES Gould, Hold Rinchart and Winston, New York.
  3. "Organic Chemistry" Clayden, Greeves, Warren and Woihers., Oxford University Press 2001.
  4. "Organic Chemistry" Vol I and II. I.L. Finar. ELBS, Pearson Education Lts, Dorling Kindersley 9 India) Pvt. Ltd.
  5. A guide to mechanisms in Organic Chemistry, Peter Skyes (Orient Longman, New Delhi).
  6. Reactive Intermediates in Organic Chemistry, Tandom and Gowel, Oxford & IBH Publishers.
  7. Combinational Chemistry – Synthesis and applications – Stephen R Wilson & Anthony W Czarnik, Wiley – Blackwell.

8. Carey, Organic Chemistry, 5th Edition (Viva Books Pvt. Ltd.)
9. Organic Synthesis - The Disconnection Approach, S. Warren, Wily India
10. Principles of Organic Synthesis, ROC Norman and JM Coxan, Nelson Thorns.
11. Organic Synthesis - Special Techniques. VK Ahluwalia and R Agarwal, Narosa Publishers.
12. Organic Reaction Mechanisms IVth Edtn, VK Ahluwalia and RK Parashar, Narosa Publishers

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS704 B: ACADEMIC SPEAKING (Sem-I)

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/ Week	Theory		Practical		Total
					Internal	External	Internal	External	
1	HS704 A/B/C/D/ E/F/G/H	Academic Speaking	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	**Foundations of Advance Communication** • *Meaning and Definition of Advance Communication* • *Advance Communication in Digital, Social, Mobile World* • *Strategies for Advance Communication* • Meaning and Concept of Academic Language • High Frequency Academic Vocabulary	**04**
2	**Art of Conversation** • *Describing people, places and things* • *Expressing opinions* • *Making suggesting* • *Persuading someone* • *Interpreting and Summarizing*	**06**

3	***Science of Power Speaking*** • *Phonemes* • *Word Stress* • *Pronunciation* • *Intonation* • *Pause* • *Register* • *Fluency* • *Prosody* • *Lexical Range*	**06**
4	***Academic Speaking Application – Part I*** • *Art of Oratory* • *Formal Presentation* • *Speech Analysis – Decoding Best Speeches*	**08**
5	**Academic Speaking Application – Part II** • *Job Interview* • *Group Discussion* • *Meeting*	**06**
	Total	**30**

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, group work, independent and collaborative research, presentations etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	I-Talk	1	10	25
2	Situational Speaking	1	05	
3	Case Study - Speech Analysis	2	10	
4	Attendance and Class Participation	-		05
			Total	**30**

External Evaluation

The University Practical Examination will be for 70 marks and will test the advance communication skills and academic speaking.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course the student would:

CO1	understand and demonstrate advance communication skills and academic speaking.
CO2	demonstrate linguistic competence
CO3	demonstrate performing ability at group discussion and personal interview.
CO4	demonstrate the formal presentation skills.
CO5	demonstrate ability to communicate in diverse situations

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	3	-	-	-
CO2	-	-	-	-	-	-	-	3	-	-	-
CO3	-	-	-	-	3	-	-	3	-	-	-
CO4	-	-	-	-	-	-	-	3	-	-	-
CO5	-	2	-	-	3	-	-	3	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books

- Headway Academic Skills - Level 1: Listening, Speaking and Study Skills Student's Book Paperback

VI. Reading

- **Unit 1:** Business communication Today (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina
- **Unit 2:** Effective Speaking Skills by Terry O' Brien
- **Unit 2:** Speak Better Write Better by Norman Lewis
- **Unit 2:** Well Spoken: Teaching Speaking to All Students by Erik Palmer
- **Unit 3:** Let Us Hear Them Speak : Developing Speaking – Listening Skills in English by Jayshree Mohanraj (Publisher – Sage Publication)
- **Unit 4:** The craft of scientific presentations: Critical steps to succeed and critical errors to avoid. New York: Springer by Michael Alley
- **Unit 4:** Presentation Skills in English by Bob Dignen (Publisher: Orient Black Swan)

FACULTY OF PHARMACY
Master of Pharmacy Programme

Syllabi
Semester II

Charotar University of Science and Technology

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM006: ADVANCE SPECTRAL ANALYSIS (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	UV and IR spectroscopy	12
2	NMR spectroscopy	12
3	Mass Spectroscopy	12
4	Chromatography	12
5	Miscellaneous analysis	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**UV and IR spectroscopy**	**12**	**20%**
	Wood ward – Fieser rule for 1,3- butadienes, cyclic dienes and α, β-carbonyl compounds and interpretation compounds of enones. ATR-IR, IR Interpretation of organic compounds.		
2.	**NMR spectroscopy**	**12**	**20%**
	1-D and 2-D NMR, NOESY and COSY, HECTOR, INADEQUATE techniques, Interpretation of organic compounds.		
3.	**Mass Spectroscopy**	**12**	**20%**
	Mass fragmentation and its rules, Fragmentation of important functional groups like alcohols, amines, carbonyl groups and alkanes, Meta stable ions, Mc Lafferty rearrangement, Ring rule, Isotopic peaks, Interpretation of organic compounds.		
4.	**Chromatography**	**12**	**20%**
	Principle, Instrumentation and Applications of the following : a)		

	GC-MS b) GC-AAS c) LC-MS d) LC-FTIR e) LC-NMR f) CE-MS g) High Performance Thin Layer chromatography h) Super critical fluid chromatography i) Ion Chromatography j) I-EC (Ion-Exclusion Chromatography) k) Flash chromatography		
5.	**Miscellaneous analysis**	**12**	**20%**
	a) Thermal methods of analysis Introduction, principle, instrumentation and application of DSC, DTA and TGA. b). Raman Spectroscopy Introduction, Principle, Instrumentation and Applications. c). Radio immuno assay Biological standardization, bioassay, ELISA, Radioimmuno assay of digitalis and insulin.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	interpret NMR, Mass and IR spectra of various organic compounds
CO2	understand theoretical and practical skills of the hyphenated instruments
CO3	identify organic compounds by spectral analysis and assay

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	3	-	-	-	-	-	-	-	3

Recommended Study Material:

- ❖ **Text book:**

1. Spectrometric Identification of Organic compounds - Robert M Silverstein, Sixth edition, John Wiley & Sons, 2004.

2. Principles of Instrumental Analysis - Doglas A Skoog, F. James Holler, Timothy A. Nieman, 5th edition, Eastern press, Bangalore, 1998.

3. Instrumental methods of analysis – Willards, 7th edition, CBS publishers.
4. Organic Spectroscopy - William Kemp, 3rd edition, ELBS, 1991.
5. Quantitative analysis of Pharmaceutical formulations by HPTLC - P D Sethi, CBS Publishers, New Delhi.
6. Quantitative Analysis of Drugs in Pharmaceutical formulation - P D Sethi, 3rd Edition, CBS Publishers, New Delhi, 1997.
7. Pharmaceutical Analysis- Modern methods – Part B - J W Munson, Volume 11, Marcel Dekker Series.

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM007: ADVANCED ORGANIC CHEMISTRY – II (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Green Chemistry	12
2	Chemistry of Peptides	12
3	Photochemical and Pericyclic Reactions	12
4	Catalysis	12
5	Stereochemistry & Asymmetric Synthesis	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Green Chemistry**	**12**	**20%**
	a. Introduction, principles of green chemistry b. Microwave assisted reactions: Merit and demerits of its use, increased reaction rates, mechanism, superheating effects of microwave, effects of solvents in microwave assisted synthesis, microwave technology in process optimization, its applications in various organic reactions and heterocycles synthesis c. Ultrasound assisted reactions: Types of sonochemical reactions, homogenous, heterogeneous liquid-liquid and liquid-solid reactions, synthetic applications d. Continuous flow reactors: Working principle, advantages and synthetic applications.		
2.	**Chemistry of Peptides**	**12**	**20%**
	a. Coupling reactions in peptide synthesis		

	b. Principles of solid phase peptide synthesis, t-BOC and FMOC protocols, various solid supports and linkers: Activation procedures, peptide bond formation, deprotection and cleavage from resin, low and high HF cleavage protocols, formation of free peptides and peptide amides, purification and case studies, site-specific chemical modifications of peptides c. Segment and sequential strategies for solution phase peptide synthesis with any two case studies d. Side reactions in peptide synthesis: Deletion peptides, side reactions initiated by proton abstraction, protonation, over activation and side reactions of individual amino acids.		
3.	**Photochemical and Pericyclic Reactions**	**12**	**20%**
	a. Basic principles of photochemical reactions. Photo-oxidation, photo-addition and photo-fragmentation. b. Mechanism, Types of pericyclic reactions such as cyclo addition, electrocyclic reaction and sigmatrophic rearrangement reactions with examples.		
4.	**Catalysis**	**12**	**20%**
	a. Types of catalysis, heterogeneous and homogenous catalysis, advantages and disadvantages. b. Heterogeneous catalysis – preparation, characterization, kinetics, supported catalysts, catalyst deactivation and regeneration, some examples of heterogeneous catalysis used in synthesis of drugs. c. Homogenous catalysis, hydrogenation, hydroformylation, hydrocyanation, Wilkinson catalysts, chiral ligands and chiral induction, Ziegler-Natta catalysts, some examples of homogenous catalysis used in synthesis of drugs. d. Transition-metal and Organo-catalysis in organic synthesis: Metal-catalyzed reactions e. Biocatalysis: Use of enzymes in organic synthesis,		

	immobilized enzymes/cells in organic reaction. f. Phase transfer catalysis - theory and applications.		
5.	**Stereochemistry & Asymmetric Synthesis**	**12**	**20%**
	Basic concepts in stereochemistry – optical activity, specific rotation, racemates and resolution of racemates, the Cahn, Ingold, Prelog (CIP) sequence rule, meso compounds, pseudo asymmetric centres, axes of symmetry, Fischers D and L notation, cis-trans isomerism, E and Z notation. Methods of asymmetric synthesis using chiral pool, chiral auxiliaries and catalytic asymmetric synthesis, enantiopure separation and Stereo selective synthesis with examples.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	describe the principles and applications of Green chemistry
CO2	understand the concept of peptide chemistry
CO3	summarize the various catalysts used in organic reactions
CO4	comprehend concept of stereochemistry and asymmetric synthesis.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ **Text book:**

1. "Advanced Organic chemistry, Reaction, mechanisms and structure", J March, John Wiley and sons, New York.

2. “Mechanism and structure in organic chemistry”, ES Gould, Hold Rinchart and Winston, New York.

3. “Organic Chemistry” Clayden, Greeves, Warren and Woihers. Oxford University Press 2001.

4. “Organic Chemistry” Vol I and II. I.L. Finar. ELBS, Sixth ed., 1995.

5. Carey, Organic chemistry, 5th edition (Viva Books Pvt. Ltd.)

6. Organic synthesis-the disconnection approach, S. Warren, Wily India

7. Principles of organic synthesis, R O C Norman and J M Coxan.

8. Organic synthesis- Special techniques VK Ahluwalia and R Aggarwal, Narosa Publishers.

9. Organic reaction mechanisms IVth edition, VK Ahluwalia and RK Parashar, Narosa Publishers.

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM008: COMPUTER AIDED DRUG DESIGN (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Quantitative Structure Activity Relationships-Basic	12
2.	Quantitative Structure Activity Relationships-Application	12
3.	Molecular Modeling and Docking	12
4.	Molecular Properties and Drug Design	12
5.	Pharmacophore Mapping and Virtual Screening	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Quantitative Structure Activity Relationships-Basic**	**12**	**20%**
	History and development of QSAR: Physicochemical parameters and methods to calculate physicochemical parameters: Hammett equation and electronic parameters (sigma), lipophilicity effects and parameters (log P, pi-substituent constant), steric effects (Taft steric and MR parameters) Experimental and theoretical approaches for the determination of these physicochemical parameters.		
2.	**Quantitative Structure Activity Relationships: Applications**	**12**	**20%**
	Hansch analysis, Free Wilson analysis and relationship between them, Advantages and disadvantages; Deriving 2D-QSAR equations. 3D-QSAR approaches and contour map analysis. Statistical methods used in QSAR analysis and importance of statistical parameters.		

3.	**Molecular Modeling and Docking**	**12**	**20%**
	a) Molecular and Quantum Mechanics in drug design. b) Energy Minimization Methods: comparison between global minimum conformation and bioactive conformation c) Molecular docking and drug receptor interactions: Rigid docking, flexible docking and extra-precision docking. Agents acting on enzymes such as DHFR, HMG-CoA reductase and HIV protease, choline esterase (AChE & BChE)		
4.	**Molecular Properties and Drug Design**	**12**	**20%**
	a) Prediction and analysis of ADMET properties of new molecules and its importance in drug design. b) De novo drug design: Receptor/enzyme-interaction and its analysis, Receptor/enzyme cavity size prediction, predicting the functional components of cavities, Fragment based drug design. c) Homology modeling and generation of 3D-structure of protein		
5.	**Pharmacophore Mapping and Virtual Screening**	**12**	**20%**
	a. Concept of pharmacophore, pharmacophore mapping, identification of Pharmacophore features and Pharmacophore modeling; Conformational search used in pharmacophore mapping b. *In Silico* Drug Design and Virtual Screening Techniques Similarity based methods and Pharmacophore based screening, structure based In-silico virtual screening protocols.		

Course Outcome (COs):

At the end of the course, the students would be able to;

Role of CADD in drug discovery • Different CADD techniques and their applications • Various strategies to design and develop new drug like molecules. • Working with molecular modeling softwares to design new drug molecules • The in silico virtual screening protocols

CO1	understand role of CADD in drug discovery.
CO2	summarize different CADD techniques and their applications
CO3	comprehend various strategies to design and develop new drug like molecules
CO4	describe in silico virtual screening protocols and molecular modeling softwares.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	3	-	-	-	-	-	-	3

Recommended Study Material:

❖ **Textbook:**

1. Computational and structural approaches to drug discovery, Robert M Stroud and Janet. F Moore, RCS Publishers.
2. Introduction to Quantitative Drug Design by Y.C. Martin, CRC Press, Taylor & Francis group.
3. Drug Design by Ariens Volume 1 to 10, Academic Press, 1975, Elsevier Publishers.
4. Principles of Drug Design by Smith and Williams, CRC Press, Taylor & Francis.
5. The Organic Chemistry of the Drug Design and Drug action by Richard B. Silverman, Elsevier Publishers.
6. Medicinal Chemistry by Burger, Wiley Publishing Co.
7. An Introduction to Medicinal Chemistry –Graham L. Patrick, Oxford University Press.
8. Wilson and Gisvold’s Text book of Organic Medicinal and Pharmaceutical Chemistry, Ippincott Williams & Wilkins.
9. Comprehensive Medicinal Chemistry – Corwin and Hansch, Pergamon Publishers.
10. Computational and structural approaches to drug design edited by Robert M Stroud and Janet. F Moore.

FACULTY OF PHARMACY

MASTER OF PHARMACY PROGRAMME

PHPCM009: PHARMACEUTICAL PROCESS CHEMISTRY (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Process Chemistry	12
2	Unit operations	12
3	Unit Processes – I	12
4	Unit Processes – II	12
5	Industrial Safety	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Process Chemistry**	**12**	**20%**
	Introduction, Synthetic strategy Stages of scale up process: Bench, pilot and large scale process. In-process control and validation of large scale process. Case studies of some scale up process of APIs. Impurities in API, types and their sources including genotoxic impurities		
2.	**Unit operations**	**12**	**20%**
	a) Extraction: Liquid equilibria, extraction with reflux, extraction with agitation, counter current extraction. b) Filtration: Theory of filtration, pressure and vacuum filtration, centrifugal filtration. c) Distillation: azeotropic and steam distillation d) Evaporation: Types of evaporators, factors affecting evaporation.		

	e) Crystallization: Crystallization from aqueous, non-aqueous solutions factors affecting crystallization, nucleation. Principle and general methods of Preparation of polymorphs, hydrates, solvates and amorphous APIs.		
3.	**Unit Processes – I**	**12**	**20%**
	a) Nitration: Nitrating agents, Aromatic nitration, kinetics and mechanism of aromatic nitration, process equipment for technical nitration, mixed acid for nitration. b) Halogenation: Kinetics of halogenations, types of halogenations, catalytic halogenations. Case study on industrial halogenation process. c) Oxidation: Introduction, types of oxidative reactions, Liquid phase oxidation with oxidizing agents. Non-metallic Oxidizing agents such as H_2O_2, sodium hypochlorite, Oxygen gas, ozonolysis.		
4.	**Unit Processes – II**	**12**	**20%**
	a) Reduction: Catalytic hydrogenation, Heterogeneous and homogeneous catalyst; Hydrogen transfer reactions, Metal hydrides. Case study on industrial reduction process. b) Fermentation: Aerobic and anaerobic fermentation. Production of i. Antibiotics; Penicillin and Streptomycin, ii. Vitamins: B2 and B12 iii. Statins: Lovastatin, Simvastatin c) Reaction progress kinetic analysis i. Streamlining reaction steps, route selection, ii. Characteristics of expedient routes, characteristics of cost-effective routes, reagent selection, families of reagents useful for scale-up.		
5.	**Industrial Safety**	**12**	**20%**
	a) MSDS (Material Safety Data Sheet), hazard labels of chemicals and Personal Protection Equipment (PPE) b) Fire hazards, types of fire & fire extinguishers c) Occupational Health & Safety Assessment Series 1800		

	(OHSAS-1800) and ISO-14001(Environmental Management System), Effluents and its management		

Course Outcome (COs):

At the end of the course, the students would be able to;

The strategies of scale up process of apis and intermediates • The various unit operations and various reactions in process chemistry

CO1	describe the strategies of scale up process of APIs and intermediates.
CO2	summarize various unit operations and various reactions in process chemistry.
CO3	comprehend hazard and safety management of chemical and equipment.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	3	3

Recommended Study Material:

❖ **Textbook:**

1. Process Chemistry in the Pharmaceutical Industry: Challenges in an Ever Changing Climate-An Overview; K. Gadamasetti, CRC Press.
2. Pharmaceutical Manufacturing Encyclopedia, 3rd edition, Volume 2.
3. Medicinal Chemistry by Burger, 6th edition, Volume 1-8.
4. W.L. McCabe, J.C Smith, Peter Harriott. Unit operations of chemical engineering, 7th edition, McGraw Hill
5. Polymorphism in Pharmaceutical Solids .Dekker Series Volume 95 Ed: H G Brittain (1999)
6. Regina M. Murphy: Introduction to Chemical Processes: Principles, Analysis, Synthesis

7. Peter J. Harrington: Pharmaceutical Process Chemistry for Synthesis: Rethinking the Routes to Scale-Up
8. P.H. Groggins: Unit processes in organic synthesis (MGH)
9. F.A. Henglein: Chemical Technology (Pergamon)
10. M. Gopal: Dryden's Outlines of Chemical Technology, WEP East-West Press
11. Clausen, Mattson: Principle of Industrial Chemistry, Wiley Publishing Co.,
12. Lowenheim & M.K. Moran: Industrial Chemicals
13. S.D. Shukla & G.N. Pandey: A text book of Chemical Technology Vol. II, Vikas Publishing House
14. J.K. Stille: Industrial Organic Chemistry (PH)
15. Shreve: Chemical Process, Mc Grawhill.
16. B.K. Sharma: Industrial Chemistry, Goel Publishing House
17. ICH Guidelines
18. United States Food and Drug Administration official website www.fda.gov

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM010: PHARMACEUTICAL CHEMISTRY PRACTICAL-II
(Practical)

Total hours: 180 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Synthesis of organic compounds by adapting different approaches involving (3 experiments) a) Oxidation b) Reduction/hydrogenation c) Nitration
2	Comparative study of synthesis of APIs/intermediates by different synthetic routes (2 experiments)
3	Assignments on regulatory requirements in API (2 experiments)
4	Comparison of absorption spectra by UV and Wood ward – Fieser rule
5	Interpretation of organic compounds by FT-IR.
6	Interpretation of organic compounds by MASS
7	Interpretation of organic compounds by NMR
8	Determination of purity by DSC in pharmaceuticals
9	Identification of organic compounds using FT-IR, NMR, CNMR and Mass spectra.
10	To carry out the preparation of following organic compounds a. Preparation of 4-chlorobenzhydrylpiperazine. (an intermediate for cetirizine HCl). b. Preparation of 4-iodotolene from p-toluidine. c $NaBH_4$ reduction of vanillin to vanillyl alcohol d. Preparation of umbelliferone by Pechhman reaction e. Preparation of triphenyl imidazole
11	To perform the Microwave irradiated reactions of synthetic importance (Any two) such as phenytoin and Biginelii compound.
12	Determination of log P, MR, hydrogen bond donors and acceptors of selected drugs using software
13	Calculation of ADMET properties of drug molecules and its analysis using software.

14	Pharmacophore modelling based experiment
15	2D-QSAR and 3D QSAR based experiments
16	Docking study and virtual screening based experiment

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	elucidate organic compound by spectroscopic techniques.
CO2	synthesize various APIs and intermediates by conventional and green chemistry approach
CO3	perform and interpret in silico ADMET properties of organic compounds
CO4	perform virtual screening and pharmacophore based experiment
CO5	analyze the problem, communicate suggested solution and interpret the results

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	3	3
CO3	3	-	3	3	-	-	-	-	-	-	3
CO4	3	-	3	3	-	-	-	-	-	-	3
CO5	3	-	3	-	-	-	-	3	-	-	3

Recommended Study Material:

- ❖ **Textbook:**

  1. "Advanced Organic chemistry, Reaction, Mechanisms and Structure", J March, John Wiley and Sons, New York.
  2. "Mechanism and Structure in Organic Chemistry", ES Gould, Hold Rinchart and Winston, New York.
  3. "Organic Chemistry" Clayden, Greeves, Warren and Woihers., Oxford University Press 2001.
  4. "Organic Chemistry" Vol I and II. I.L. Finar. ELBS, Pearson Education Lts, Dorling Kindersley 9 India) Pvt. Ltd.
  5. A guide to mechanisms in Organic Chemistry, Peter Skyes (Orient Longman, New Delhi).
  6. Reactive Intermediates in Organic Chemistry, Tandom and Gowel, Oxford & IBH Publishers.
  7. Combinational Chemistry – Synthesis and applications – Stephen R Wilson & Anthony W Czarnik, Wiley – Blackwell.

8. Carey, Organic Chemistry, 5th Edition (Viva Books Pvt. Ltd.)
9. Organic Synthesis - The Disconnection Approach, S. Warren, Wily India
10. Principles of Organic Synthesis, ROC Norman and JM Coxan, Nelson Thorns.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS705 (B): ACADEMIC WRITING

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
II	HS705 (B)	Academic Writing	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title / Topic	Classroom Contact Hours
1	**Academic Writing and Research Process** • *Introduction to Academic Writing* • *Academic Writing as a Part of Research* • *Types of Academic Writing* • *Features of Academic Writing* • *Importance of Good Academic Writing in various Academic Works*	**05**
2	**Anatomy of Academic Writing** • *Academic Vocabulary* • *Simple and Complex Sentences* • *Organizing Paragraphs* • *The Writing Process* • *Adopting Academic Writing Style*	**05**
3	**Key Academic Skills** • Note – taking	**05**

	• Note – making • Paraphrasing • Summarizing	
4	**Accuracy in Academic Writing** • *Lexical Range* • *Academic Language and Structures* • *Elements of Writing* • *Proof Reading, Editing, and Rewriting*	**05**
5	**Using and Citing Sources of Ideas** • *Academic Texts and their Types* • *Intellectual Honesty in Academic Writing* • *Avoiding Plagiarism – Idea Theft* • *Degrees of Plagiarism* • *Types of Borrowing* • *Anatomy of Citations* • *Common Citation Styles*	**05**
6	**Contemporary Practices in Academic Writing** • Analytical Essays • Graph / Table / Process Interpretation and Description • Writing Reports • Writing Research / Concept Papers	**05**
Total		**30**

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like writing, group work, independent and collaborative research, etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Paragraph Writing	1	3	03
2	Note-taking / Note-making	1	3	03
3	Paraphrasing / Summarizing	1	4	04
4	Essay Writing	1	5	05
5	Concept Paper Writing	1	10	10
5	Attendance and Class Participation			05
			Total	**30**

External Evaluation

The University Practical Examination will be for 70 marks and will test the professional communication skills and academic writing skills of the students.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical /Quiz/ Project / Academic Writing	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course, the student would:

CO1	have sound understanding of the concept and applications of academic writing
CO2	have acquired enough knowledge of academic writing style, strategy and approach
CO3	be able to demonstrate error free and effective academic writing
CO4	be able to demonstrate ability to work on project/report/paper writing
CO5	understand the concept of plagiarism and learn to use different citation styles as a part of referencing

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	3	-	-	-
CO2	-	-	-	-	-	-	-	3	-	-	-
CO3	-	-	-	-	-	-	-	3	-	-	-
CO4	-	-	-	-	-	-	-	3	-	-	-
CO5	-	-	-	-	-	-	2	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books / Reading

Essential Reading for Concepts

- Academic Writing for International Students, Routledge
- Academic Writing: A Guide for Management Students and Researchers. Monipally, M. M. & Pawar, B. S. Sage. 2010. New Delhi

Essential Reading for Activity and Teacher Resource

- ***Effective Academic Writing Level - 1,2,3,4 (Second Edition) By:*** Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; *Publisher: Oxford*

Additional Reading

- Writing Your Thesis (2^{nd} Edition) by Paul Oliver, Sage
- Development Communication In Practice by Vilanilam V J, Sage
- Intercultural Communication by Mingsheng Li, Patel Fay, Sage
- www.owl.perdue.edu.

FACULTY OF PHARMACY
Master of Pharmacy Programme

Syllabi
Semester III

Charotar University of Science and Technology

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHCCC010 : RESEARCH METHODOLOGY & BIOSTATISTICS

Total hours: 60

UNIT – I General Research Methodology: **12 hr**

Research, objective, requirements, practical difficulties, review of literature, study design, types of studies, strategies to eliminate errors/bias, controls, randomization, crossover design, placebo, blinding techniques.

UNIT – II Biostatistics: **12 hr**

Definition, application, sample size, importance of sample size, factors influencing sample size, dropouts, statistical tests of significance, type of significance tests, parametric tests(students "t" test, ANOVA, Correlation coefficient, regression), non-parametric tests (wilcoxan rank tests, analysis of variance, correlation, chi square test), null hypothesis, P values, degree of freedom, interpretation of P values.

UNIT – III Medical Research: 12 hr

History, values in medical ethics, autonomy, beneficence, non-maleficence, double effect, conflicts between autonomy and beneficence/non-maleficence, euthanasia, informed consent, confidentiality, criticisms of orthodox medical ethics, importance of communication, control resolution, guidelines, ethics committees, cultural concerns, truth telling, online business practices, conflicts of interest, referral, vendor relationships, treatment of family members, sexual relationships, fatality.

UNIT – IV CPCSEA guidelines for laboratory animal facility: 12 hr

Goals, veterinary care, quarantine, surveillance, diagnosis, treatment and control of disease, personal hygiene, location of animal facilities to laboratories, anesthesia, euthanasia, physical facilities, environment, animal husbandry, record keeping, SOPs, personnel and training, transport of lab animals.

UNIT – V Declaration of Helsinki: 12 hr

History, introduction, basic principles for all medical research, and additional principles for medical research combined with medical care.

A. Recommended Study Material:

- **Text Books:**
  1. Research Methodology, Methods & Techniques, C.R. Kothari, Viswa Prakashan, 2nd Edition, 2009.
  2. Research Methods- A Process of Inquiry, Graziano, A.M., Raulin, M.L, Pearson Publications, 7th Edition, 2009.
  3. How to Write a Thesis:, Murray, R. Tata McGraw Hill, 2nd Edition, 2010.
  4. Writing For Academic Journals, Murray, R., McGraw Hill International, 2009.
  5. Writing for Publication, Henson, K.T., Allyn &Bacon, 2005.
- **Reference Books:**
  1. What is this thing called Science, Chalmers, A.F., Queensland University Press, 1999.
  2. Methods &Techniques of Social Research, Bhandarkar & Wilkinson, Himalaya publications, 2009.
  3. Doing your Research project, Bell J., Open University Press, Berkshire, 4thEdition, 2005
  4. A Handbook of Academic Writing, Murray, R. and Moore, S., Tata McGraw Hill International, 2006.

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM013: RESEARCH WORK-I

Total hours: 480

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	define and describe research problem.
CO2	illustrate project management skills such as project design, scientific information and literature access, project implementation, data analysis, and interpretation.
CO3	present a dissertation report integrating appropriate written and verbal communicative skills.
CO4	efficiently use communication and information technology tools.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	3	3	3	-	-	-	-	-	-	3
CO3	3	-	3	3	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	3	3	-	-	3

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHCCC005: JOURNAL CLUB -I

Total hours: 30

Course Outcome (COs):

At the end of the course, the students would be able to

	CO
CO1	present scientific literature and interpret the finding

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	3	-	-	-	3	-	-	3

FACULTY OF PHARMACY
Master of Pharmacy Programme

Syllabi
Semester IV

Charotar University of Science and Technology

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHPCM015 : RESEARCH WORK-II

Total hours: 480

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	define and describe research problem.
CO2	illustrate project management skills such as project design, scientific information and literature access, project implementation, data analysis, and interpretation.
CO3	present a dissertation report integrating appropriate written and verbal communicative skills.
CO4	efficiently use communication and information technology tools.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	3	3	3	-	-	-	-	-	-	3
CO3	3	-	3	3	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	3	3	-	-	3

FACULTY OF PHARMACY
MASTER OF PHARMACY PROGRAMME
PHCCC006: JOURNAL CLUB -II

Total hours: 30

Course Outcome (COs):

At the end of the course, the students would be able to

	CO
CO1	present scientific literature and interpret the finding

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	3	-	-	-	3	-	-	3

UNIVERSITY ELECTIVE LEVEL-POSTGRADUATE

CHAROTAR UNIVERSITY OF SCIENCE&TECHNOLOGY

List of University Elective Courses

First Semester Postgraduate Programme

(Academic year 2019-20)

Sr. No.	Course Code & Course Name	Department / Faculty offering the course
1	MA771.01: Reliability and Risk Analysis-I	PDPIAS / FAS
2	EE781.01:Optimization Techniques	EE/FTE
3	CE772.01:Research Methodology	CE / FTE
4	CA730: Internet and Web Designing	CMPICA / FCA
5	PT795.01 Heath and Physical activity	ARIP / FMD
6	NR755: First aid and Life support	NURSING / FMD
7	RD701.01: Introduction to analytical Techniques	PDPIAS/FAS
8	RD702.01:Introduction to Nanoscience and Technology	PDPIAS/FAS
9	MB650: Creative Leadership	I2IM / FMS
10	PH891:Community Pharmacy ownership	RPCP/FPH
11	ME781.01: Occupational Health and Safety	ME/FTE
12	PD261: Astrophysics, Space and Cosmos-1(ASC-1)	PDPIAS/FAS

MA771.01: RELIABILITY AND RISK ANALYSIS-I

Total Hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of the course:

Sr No.	Title of the unit	Minimum number of hours
1.	Introduction to Probability and Statistics	06
2.	Characteristics of Reliability	06
3.	Reliability of Simple Systems	06
4.	Concept of Safety and Risk Analysis	06
5.	Reliability Modeling	06

B. Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1	**Introduction to Probability and Statistics**	**06 Hours**	**20%**
1.1	Random Event.		
1.2	Basic formula of Probability.		
1.3	Random Variable and Probability Distribution Functions.		
1.4	descriptive statistics		
2	**Characteristics of Reliability**	**06 Hours**	**20%**
2.1	Reliability of a Unit Functioning until First Failure		
2.2	System Reliability		
2.3	Testing for Reliability		
2.4	Exponential Law and Evaluation of parameter		

3.	**Reliability of Simple Systems**	**06 Hours**	**20%**
3.1	Series System		
3.2	Parallel System		
3.3	K out of N systems		
4.	**Concept of Safety and Risk Analysis**	**06 Hours**	**20%**
4.1	Qualitative definition of Risk		
4.2	Quantitative definition of Risk		
4.3	Failure Model and Effect Analysis(FMEA)		
4.4	Hazard and operability analysis(HAZOP)		
4.5	Fault Tree Analysis		
5.	**Reliability Modeling**	**06 Hours**	**20%**
5.1	Software Reliability Analysis		
5.2	Human Reliability		
5.3	Stress-Strength Analysis		

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures/laboratory which carries a 5% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weighting of 5%.
- Two Quizzes (surprise test) will be conducted which carries 5% component of the overall evaluation.

D. Student Learning Outcomes:

- At the end of the course the students will be able to understand the basic concepts of Reliability and Risk Analysis.

- Student will be able to apply concepts of these course in their study of specialization

E. Recommended Study Material:

- **Text Books:**

1. Mathematical Methods of Reliability Theory. B. V. Gnedenko, Yu. K. Belyayev, and A. D. Solovyev ,Academic Press 1969
2. An Introduction to Basics of Reliability and Risk Analysis. Enric Zio, World Scientific Publishing Co.Pte. Ltd.2007
3. Reliability and Risk Analysis. Terje Aven, Elsevier Publicaion,1992

- **Reference Books/Articles:**

1. On The Quantitative Definition of Risk, Stanley Kaplan and B. John Garrick, Risk Analysis, Vol. I, No. I , 1981
2. Probability concepts in engineering planning and design. Volume II – decision, risk and reliability. Ang, A.H.-S. and Tang, W.H John Wiley & Sons, Inc., New York (1984)
3. M. Modarres, Reliability and Risk Analysis, Marcel Dekker (1993).
4. N.J. McCormick, Reliability and Risk Analysis, Academic Press (1981).

EE781.01: OPTIMIZATION TECHNIQUES

Total Hours (Theory): 30

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of the Course:

Sr. No.	Title of Unit	Min. no of hours
1	Fundamentals of Optimization	01
2	Linear Programming	05
3	Unconstrained Optimization	04
4	Nonlinear Programming	05
5	Fundamentals of Artificial Intelligence methods	01
6	Particle Swarm Optimization and Cuckoo Search Algorithm	08
7	MATLAB programming and Optimization Toolbox	06

B. Detailed Syllabus

Sr. No.	UNIT	Hours	Weightage (%)
1	**Fundamentals of Optimization**	01 Hours	3.33%
	Introduction, Feasibility and optimality, Convexity, constraints, Rates of convergence		
2	**Linear programming**	05 Hours	16.66%
	Introduction, Formulation of Linear programming problem, Graphical method, Simplex method, Basic solution, Basic feasible solution, Simplex algorithm, Two phase method.		
3	**Unconstrained Optimization**	04 Hours	13.33%
	Introduction, Optimality conditions, Newton's method for minimization, Line search methods, Steepest-Descent method, Quasi-Newton method, Modified newton's method		
4	**Nonlinear Programming**	05 Hours	16.66%

	Optimality conditions for constrained problems, Kuhn-Tucker conditions, Penalty function method, Barrier method, The Lagrange multipliers and the Lagrangian function, Sensitivity analysis, Computing the Lagrange multipliers, Sequential quadratic programming, Interior point method		
5	**Fundamentals of Artificial Intelligence methods**	01 Hours	3.33%
	To understand importance of AI methods and their comparison with various classical methods using various criteria.		
6	**Particle Swarm Optimization and Cuckoo Search Algorithm**	08 Hours	26.66%
	Introduction to PSO, Unconstrained and constrained optimization using PSO, Effects of various coefficients on convergence, Cuckoo Search (CS) Algorithm, Comparison between PSO and CS		
7	**MATLAB programming and Optimization Toolbox**	06 Hours	20%
	MATLAB programming of various classical and AI methods. Use of MATLAB optimization toolbox to solve various optimization problems.		

C. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a component of the overall evaluation.
- Minimum two internal exams will be conducted and will be considered as a part of overall evaluation.
- Assignments based on course content will be given to the students for each unit/topic and will be evaluated at regular interval and its weightage may be reflected in the overall evaluation.

D. Student Learning Outcomes:

A. The students will be able to get awareness about the optimization problems. They can differentiate the class of classical optimization methods and AI methods

B. The student will learn to handle, solve and analyzing problems using linear programming and other mathematical programming algorithms

C. The students will also be able to learn different techniques to solve Non- Linear Programming Problems. They can also use search techniques methods, which are based on iterative methods, to find optimal solutions of Non-Linear Programming Problems.

D. Ability to develop codes for evolutionary optimization techniques and to solve wide range of optimization problem.

E. The students will be able to solve optimization methods using software tools such as MATLAB and be prepared for developing case studies and simulation examples

E. Recommended Study Material:

- **Books:**

1. "Optimization Methods for Engineers", N.V.S. Raju, PHI, 2014.
2. "Artificial intelligence and intelligent systems", N.P. Padhy, Oxford University Press, 2005
3. "Engineering Optimization: Theory and Practice", S. Rao, 4th Edition, John Wiley & Sons, Inc., 2009

CE772.01: RESEARCH METHODOLOGY

Total hours: 30

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	General introduction to Research	02
2	Research problem Formulation	03
3	Research Design	08
4	Research Publication & Presentation	08
5	Research Ethics and Morals	05
6	Quality indices of research publication	04

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1	**General introduction to Research**	**02 Hours**	**06%**
	General Introduction: Importance of Research, Role of Research, Aims & Objectives, Research Process, Phases of Research. Introduction to Research Methodology: Meaning of Research, Objectives of Research, Motivations in Research, Types of Research, Research Approaches, Significance of Research, Research Methods v/s Methodology, Research and Scientific Methods, Research Process, Criteria of Good Research.		
2.	**Research problem Formulation:**	**03 Hours**	**09%**
	Review of Research Literature: Defining the Research Problem: What is Research Problem, Selecting the		

	Problem, Necessity of and Techniques in defining the problem. Purpose and use of literature review, locating relevant information, use of library & electronic databases, preparation & presentation of literature review, research article reviews, theoretical models and frame work. Identification of gaps in research, formulation of research problem, definition of research objectives.		
3.	**Research Design:**	**08 Hours**	**27%**
	Research Design: Meaning, Need, Features of Good Design, Concepts, Types. Basic Principles of Experimental Design, Developing a Research Plan. *Qualitative Methods:* Types of hypothesis and characterization. *Quantitative Methods:* Statistical methods for testing and evaluation. *Characterization of experiments:* Accuracy, reliability, reproducibility, sensitivity, Documentation of ongoing research. *Sample Design:* Implication, Steps. Criteria for selecting a sample procedure, Characteristics of Good sampling Procedure, Types of Sample Design, Selecting Random Samples, Complex random sampling Design. *Measurement and Scaling Techniques:* Measurement in Research, Measurement Scales, Sources of Errors in measurement, Tests of Second measurement, Technique of developing Measurement Tools, Meaning of Scaling, Scale Classification Bases, Important Scaling Techniques, Scale Construction Techniques. *Methods of Data Collection:* Collection of Primary Data, Observation Method, Interview method, Collection of Data through questionnaire and Schedules, Other methods. Collection of Secondary Data, Selection of appropriate method for data collection, Case Study Method, Guidelines for developing questionnaire, successful interviewing. Survey v/s Experiment. *Processing and Analysis of Data :* Processing Operations (Meaning, Problems), Data Analysis (Elements), Statistics in Research, Measures of Central Tendency, Dispersion, Asymmetry, Relationship. Regression Analysis, Multiple correlation and Regression, Partial Correlation, Association in case of Attributes.		

	Sampling Fundamentals: Definition, Need, Important sampling Distribution, Central limit theorem Sampling Theory, Sandler's A-test, Concept of Standard Error, Estimation, Estimating population mean, proportion. Sample size and its determination, Determination of sample size. *Analysis of Variance and Covariance:* Basic Principles, techniques, applications, Assumptions, limitations. *Analysis of Non-parametric or distribution-free Tests :* Sign Test, Fisher-Irwin Test, McNemer Test, Wilcoxon Matched pair Test (Signed Rank Test). *Sum Tests :* a) Wilcoxon-Mann-Whitney Test b)Kruskal-Wallis Test, One sample Runs Test, Multivariate Analysis Techniques: Characteristics, Application, Classification, Variables, Techniques, Factor Analysis (Methods, Rotation), Path Analysis.		
4.	**Research Publication & Presentation:**	**08 Hours**	**27%**
	Thesis, Research paper, Organization of thesis and reports, formatting issues, citation methods, references, effective oral presentation of research, Documentation of ongoing research.		
5.	**Research Ethics and Morals:**	**05 Hours**	**19%**
	Issues related to plagiarism and ethics. Intellectual Property Rights: Copy rights, Patents, Industrial Designs, Trademarks.		
6.	**Quality indices of research publication:**	**04 Hours**	**12%**
	Impact factor, Immediacy factor.		

A. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- Tutorials related to course content will be given to students.
- In the lectures discipline and behaviour will be observed strictly.

- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

B. Student Learning Outcomes:

- Research Methodology as a subject should help researchers to prepare the literature in chronological pattern and should logically analyse the concerns.
- This subject should help in framing the research problems to enhance the scale of understanding.
- In this world of global village, research papers are available in abundance; one thesis submitted by a scholar, in no way should be a repetition of a work already done.
- This subject should help researchers to use tools, techniques, concepts and world's best practices to present a unique research.

C. Recommended Study Material:

❖ **Text Books:**

1. Research Methodology, Methods & Techniques, C.R. Kothari, Viswa Prakashan, 2nd Edition, 2009.
2. Research Methods- A Process of Inquiry, Graziano, A.M., Raulin, M.L, Pearson Publications, 7th Edition, 2009.
3. How to Write a Thesis:, Murray, R. Tata McGraw Hill, 2nd Edition, 2010.
4. Writing For Academic Journals, Murray, R., McGraw Hill International, 2009.
5. Writing for Publication, Henson, K.T., Allyn &Bacon, 2005.

❖ **Reference Books:**

1. What is this thing called Science, Chalmers, A.F., Queensland University Press, 1999.
2. Methods &Techniques of Social Research, Bhandarkar & Wilkinson, Himalaya publications, 2009.
3. Doing your Research project, Bell J., Open University Press, Berkshire, 4thEdition, 2005
4. A Handbook of Academic Writing, Murray, R. and Moore, S., Tata McGraw Hill International, 2006.

CA730: INTERNET AND WEB DESIGNING

Total Hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	**2**	2	2
Marks	**NA**	100	100	

Outline of the Course:

Sr. No.	Content
1.	Overview of Internet and WWW, Basic elements of the Internet, Internet services, Internet Browsers and Servers, Hardware and Software requirements to connect to the internet, Internet Service Provider (ISP), Introduction to Internet Protocols
2.	Introduction to Web Page, Web Site, Web Browser, Overview of HTML, Structure of HTML Documents
3.	HTML Basics Tags and HTML elements
4.	List, Marquee & Hyperlink in HTML
5.	Images and Tables in HTML
6.	Forms in HTML
7.	Media Element in HTML5
8.	Introduction to Cascading Style Sheet (CSS), Ways to embed CSS in HTML
9.	CSS selectors & Layout
10.	CSS Properties
11.	Creation of Menu with CSS
12.	Introduction to Web Publishing or Hosting : Domain Name, Web Server, Website Parking, Publishing Website through FTP

❖ **Text Books:**

1. Harley Hahn: The Internet Complete Reference, 2^{nd} Edition, Tata McGraw-HILL Edition.
2. Matthew MacDonald: HTML5: The Missing Manual, O'Reilly Media, August 2011.

3. Peter Gasston: The Book of CSS3: A Developer's Guide to the Future of Web Design, No Starch Press, April 2011.

4. Richard York: Beginning CSS: Cascading Style sheets for Web Design, Wrox Press (Wiley Publishing), 2005.

- **Reference Books**:

1. Ivan Bayross: Web Enabled Commercial Application Development using HTML, JavaScript, DHTML and PHP, 4th revised edition, BPB Publication.

2. Adrian Farrel: The Internet and its Protocol – A comparative approach, Morgan Kaufmann Publishers.

3. David Mc Farland: CSS: The Missing Manual, O'Reilly, 2006.

- **Reference Links**:

1. http://www.w3schools.com [lecture notes]

2. http://www.whatwg.org/specs/web-apps/current-work/multipage/#auto-toc-4 [HTML Materials]

3. http://people.cs.pitt.edu/~mehmud/cs134-2084/lectures.html [CSS notes]

PT795.01: HEALTH & PHYSICAL ACTIVITY

Total Hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to health	10
2.	Physical Activity	10
3.	Introduction to Yoga	10

B. Detailed Syllabus:

Sr. No.	Unit	Minimum number of hours
1	**Introduction to Health**	**10 hours**
1.1	What is health?	
1.2	Healthcare delivery system: Developing Countries, Developed Countries	
1.3	Human anatomy, physiology & physical fitness	
1.4	Basics of Nutrition	
1.5	Life style disorders – obesity & diabetes	
2	**Physical Activity**	**10 hours**
2.1	What is Physical activity, exercise, physical fitness, epidemiology?	
2.2	Measurement of Physical Activity in individuals	
2.3	Physical Activity – Theoretical Perspective: Self-determination, trans theoretical	
2.4	Physical Activity and mental health – Body image, depression, problem with exercise	
2.5	Barriers & Facilitators of Physical Activity	

3	**Introduction to Yoga**	**10 hours**
3.1	What is yoga?	
3.2	Types of yoga	
3.3	Benefits of yoga to various body systems	
3.4	Asanas, Pranayam	
3.5	Yoga therapy for various back pain, asthma, stress, hypertension, diabetes	

C. Instructional Method and Pedagogy:

- Interactive classroom sessions using black-board and audio-visual aids.
- Using the available technology and resources for e-learning.
- Students will be encouraged towards self-learning and under direct interaction with course faculty.
- Students will be enabled for continuous evaluation.
- Case study, didactic mode of group discussions

D. Student Learning Outcomes:

Upon completion of the course, the student should be able to:

- Appraise the importance of exercise in maintenance of health and fitness.
- Objectively define health and physical activity in realistic environment.

E. Recommended Study Material:

❖ **Textbooks:**

1. ACSM"s "Health Related Physical Fitness Assessment Manual Lippincott Williams and Walkins USA, 2005.
2. Nilima Patel (2008) Yoga and Rehabilitation, Jaypee Publication, India

❖ **Reference books:**

1. Biddle, S. J. H., & Mutrie, N. (2008). Psychology of physical activity. London: Routledge
2. B.C. Rai. Health Education and Hygiene Published by Prakashan Kendra
3. Puri. K. Chandra. S.S. (2005). Health and Physical Education. New Delhi: Surjeet Publications

NR755: First Aid and Life Support

Total Hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of the Course:

Unit No.	Title of Unit	Prescribed Hours
I.	**Introduction and Basics of First Aid:** □ ● Rescuer Duties, Victim and Rescuer Safety ● Looking for Help □ ● After the emergency	2
II.	**Medical emergencies and their first aid:** □ ● Breathing Problems □ ● Choking in an Adult □ ● Allergic Reactions □ ● Heart Attack ● Fainting □ ● Diabetes and Low Blood Sugar □ ● Stroke □ ● Shock Abdominal maneuver, CPR, ventilation etc.	12
III.	**Injuries emergencies and their first aid:** ● Bleeding: You Can See/ You Can't See ● Wounds □ ● Burns and Electrical Injuries □ ● Fractures Bandaging, immobilization, transferring etc.	8

IV.	**Environmental Emergencies and first aid:** ● Bites and Stings ● Poison Emergencies ● Heat-Related Emergencies ● Cold-Related Emergencies	7
V.	**Preparation of First Aid Kit**	1
	Total	**30 Hours**

B. Instruction Method and Pedagogy

The course is based on theory & practical learning. Teaching will be facilitated by reading material, discussion, microteaching, task-based learning, assignments, field visit and various interpersonal activities like group work, independent and collaborative research, presentations etc. Practical will be facilitated by demonstrations and preparing check-lists etc.

C. Evaluation:

The students will be evaluated continuously in the form of internal as well as external examinations. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sr. No.	Component	Number	Marks per incidence	Total Marks
1	Assignments	1	8	8
2	Internal Test/ Model Exam	1	12	12
3	Attendance and Class Participation	Minimum 80% attendance		10
			Total	**30**

External Evaluation

The University Practical examination will be of 70 marks and will test the logic and critical thinking skills of the students by asking them to demonstrate or putting scenario based application of knowledge l avoid, as far as possible, grammatical errors and will focus on applications.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical cum Viva-Voce	01	70	70
			Total	70

D. Learning Outcomes: At the end of the course, learners will be able to:

- Demonstrate bandaging and immobilization of patient □
- Demonstrate the skill for transferring injured person □
- Demonstrate skills to clear airway & ventilate the patient □
- Demonstrate skills to perform CPR

RD701.01: INTRODUCTION TO ANALYTICAL TECHNIQUES

Total Hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. **Outline of the Course**

Sr. No.	Title of Unit	No. of hrs.
1.	HPLC	08
2.	TGA-DSC	09
3.	DLS	05
4.	PCR	08

Syllabus Topics: **Total Hours (Theory): 30**

Sr. No.	Title of Unit	Topics
1.	High Performance Liquid Chromatography (HPLC)	Introduction to HPLC, Basic Principle of HPLC, Instrumentation for HPLC, Types of Detector used in HPLC, Column efficiency in liquid chromatography
2.	TGA-DSC	General Discussion, Thermogravemetry Analysis, Instruments available for Thermogravimetric analysis, Detailed Discussion, Principle and Applications in various fields of Science and Engineering. Data Analysis
3.	DLS	Concept of Dynamic light scattering and basics of Particle size analyzer, Data Analysis and applications of DLS
4.	PCR	Introduction and principle of Polymerase Chain Reaction (PCR), Principle of PCR, Primer designing, Detailed

		methodology of PCR, Modifications of PCR, Applications of PCR

C. Instructional Methods and Pedagogy:

The topics will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Unit tests will be conducted regularly as a part of continuous evaluation and suggestions will be given to student in order to improve their performance.

D. Student Learning Outcomes / objectives:

- The Programme aims at providing students with the methodological concepts and tools needed to acquire top-level skills in the field of some selected instrumentation
- At the end students would gain experience in using these tools and analyzing the data.

E. Recommended Study Material:

❖ **Text books/Reference books**

1. Instrumental methods of analysis by Williard Merritt Dean Settle, 7th Ed. CBS publishers and distributors Pvt. Ltd.,
2. Instrumental methods of analysis by Williard Merritt Dean Settle, 7 th Ed. CBS publishers and distributors Pvt. Ltd.,
3. Principles of Gene Manipulation and Genomics by Sandy B. Primrose, Richard Twyman 7 th Ed. Wiley-Blackwell
4. Molecular Cloning: A Laboratory Manual by Joseph Sambrook, David William Russell 3 rd Ed.CSHL Press
5. Principles and Techniques of Biochemistry and Molecular Biology by Wilson and Walker 7 th Ed. Cambridge University Press
6. Principles of Instrumental Analysis by Douglas A. Skoog, F.James Holler and Timothy A.Nieman. Publisher: Saunders College Publishing.

RD 702.01: INTRODUCTION TO NANOSCIENCE & TECHNOLOGY

Total Hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of the Course:

Sr No.	Title of Unit	Minimum No. of Hrs
1.	Nanotechnology – development history & Implications of nanotechnology	6
2.	Overview on characterization and synthesis of nanostructure materials	10
3.	Overview of nanostructures & Nano devices	10
4.	New fields of nanotechnology	4

B. Syllabus Topics:

1. **Nanotechnology – development history & Implications of nanotechnology**

   Concept of nanoscience and nanotechnology, Nanotechnology in the history and in nature. Impact of the nanotechnology in the society: Ethical, social, economic and environmental implications. the nanoscale. Size dependent physical and chemical properties. Surface effects. Importance of the surface at nanoscale. The surface/volume ratio. Size dependent properties

2. **Overview on synthesis and characterization of nanostructure materials.**

   Physical, chemical and biological methods of synthesis of nanostructures, Electron microscopy, scanning probe microscopy, non-imaging techniques

3. **Overview of nanostructures & Nano devices**

   zero dimensional, one dimensional and two dimensional nanostructures , Electronic devices, magnetic devices, photonic devices, mechanical devices, fluidics devices and biomedical devices

4. New fields of nanotechnology

quantum computing, spintronics, nanomedicines, energy, etc.

C. Instructional Methods and Pedagogy:

The topics will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Students will be exposed to practical operations, and lab visit and experimental demonstrations of some of the equipment facilities for Nano fabrication & characterization available in UNI. Students will produce a technical report on the experiences.

D. Student Learning Outcomes / objectives:

Nanotechnology promises to be the technology of the future benefitting the humanity in a number of ways. This course is aimed at preparing students for further industrial or academic work in the field of nano-characterization techniques.

E. Recommended Study Material:

❖ **Text Books / Reference Books:**

1. Essentials of nanotechnology by Jeremy Ramsden [JR], 2009, Jeremy Ramsden & Ventus Publishing ApS,
2. Introduction to Nanoscience, S.M.Lindsay, Oxford ISBN 978-019-954421-9 (2010).
3. Guozhong Cao (2004). *Nanostructures and Nanomaterials: Synthesis, Properties & Applications*, 448 pages, Imperial College Press, ISBN-10: 1860944159.
4. NANO: The Essentials Understanding Nanoscience and Nanotechnology, T. Pradeep, Tata McGraw-Hill Publishing Co. Ltd., 2007.
5. Nanoscience and Nanotechnology, B K Parthsarathy, ISHA Books, New Delhi, 2007.
6. Nanotechnology: Principles and practices, Sulabha K Kulkarni, Capital publishing company, 2007.

MB650: CREATIVE LEADERSHIP

Total Hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of the Course:

Module No.	Title/Topic	Classroom Contact Sessions
1	**Introduction to Leadership** • What are Leadership Skills? • Ways of Conceptualizing Leadership • Definition and Components • A Born Leader • Traits of Successful Leader • Why Leadership? o Managerial Roles • Importance of Leadership **Leading Vs Managing** • Roles and Relationships • Developing Personality for Effective Leading Roles • Authority Vs. Responsibility • Leading the Team • Leadership–Styles, Models and Philosophy	**05**
2	**Leadership Approach** • Trait Approach • Skills Approach • Style Approach	**05**

Module No.	Title/Topic	Classroom Contact Sessions
	• Situational Approach • Psychodynamic Approach **Leadership Theories** • Contingency Theory • Path-Goal Theory • Leader-Member Exchange Theory	
3	**Leadership Processes** • Transactional Leadership • Transformational Leadership • Authentic Leadership • Team Leadership • Integrative Leadership • Liquid Leadership	**05**
4	**Women and Leadership** • Gender and Leadership Styles • Gender and Leadership Effectiveness • Glass Ceiling Turned Labyrinth • Strengths of Women Leadership • Criticism and Application	**05**
5	**Culture and Leadership** • Dimensions of Culture • Clusters of World Cultures • Leadership Behavior and Culture Clusters • Universally Desirable and Undesirable Leadership Attributes • Criticism and Application • Leadership for High Performing Organisations • General Principles for Creative Culture	**05**

Module No.	Title/Topic	Classroom Contact Sessions
	• Nurturing Personal Creativity	
6	**Contemporary Issues in Leadership** • Power and Politics in Leadership • Ethics in Leadership • Cases in Leadership	**05**
Total		**30**

B. Pedagogy

The course will emphasise self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will be as follows:

- Classroom Contact Sessions … About 20 Sessions
- Case Discussions … About 03 Sessions
- Presentation … About 03 Sessions
- Management Exercise/ Stimulations/Game … About 02 Sessions
- Feedback … About 02 Sessions

The exact division among the above components will be announced by the instructor at the beginning of the semester as a part of detailed session-wise schedule.

C. Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sr. No.	Component	Number	Marks per incidence	Total Marks	Percentage of total internal evaluation
1	Quizzes	3	10	30	10
2	Case Analysis and Presentation	2	45	90	30
3	Assignment / Project work	1	60	60	20
4	Internal Tests	2	45	90	30
5	Attendance and Class Participation			30	10
			Total	**300**	**100**

The total marks will be divided by 10 and declared as Institute-level evaluation marks for the course. The Institute-level evaluation will constitute 30% of the total marks for the course.

D. External Evaluation

The University examination will be based on oral presentation, review of students' reports and a viva-voce and will carry 70% marks for the course evaluation.

E. Learning Outcomes

At the end of the course, the student should have developed:

- Appreciation for types, traits, approaches and leadership models/theories.
- Motivation for leadership roles and responsibilities.
- Qualities of creative leadership skills.

F. Reference Material

❖ **Text-book:**

1. Leadership – Theory and Practices , Peter G. Northouse, Sage Publications India Pvt. Ltd., Latest Edition

❖ **Reference Books:**

1. Liquid Leadership by Brad Szollose, Prolibris Publishing Media, Latest Edition
2. Effective Leadership by Lussier/ Achua , Cengage Learning Publications, Latest Edition
3. Integrative Leadership by Hatala & Hatala, Pearson Power Publication, Latest Edition

4. Cases in Leadership by Rowe and Guerrero, Sage Publications India Pvt. Ltd., Latest Edition

❖ **Journals / Magazines / Newspapers:**

1. HBR Issues on Building Leadership Skills
2. International Journal of Innovation, Creativity and Change
3. Economic Times
4. Business Standard

PH891: COMMUNITY PHARMACY OWNERSHIP

Total Hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of course:

Sr. No.	Title of Unit	No. of Contact Hours
1	Introduction to Community pharmacy	10
2	Community Pharmacy Management	10
3	Pharmacy Business Plan	10
	Total	30

B. Detailed Syllabus:

Sr. No.	Title of Unit	Topics
1	Introduction to Community pharmacy	Roles & Responsibility, relationship with other health care providers, Prescribed medication order interpretation including OTC medicines, Safe use of medical devices.
2	Community Pharmacy Management	Role, process and scope of Community Pharmacy Management, modern technologies, financial, material, staff management and Drug store management, Code of ethics for Pharmacy.
3	Pharmacy Business Plan	Creating a Successful Pharmacy Business Plan, Business Owner Roles, Responsibilities, and Management Styles, Legal, Financial and Accounting Advice for the Beginning Owner, Marketing of Pharmacy Practice.

C. Instructional Methods and Pedagogy:

The content of the syllabus would be transmitted through different pedagogy tools like interactive class room sessions using classical chalk - board teaching to Power point presentations. Class room teaching would also be supplemented with group discussions, seminars, assignments and case studies.

D. Student Learning Outcomes / Objectives

At the end of the course, the student will be able to understand

- The concept of Community Pharmacy Ownership and its scope
- Students understand the role of the pharmacist in community and development
- Able to learn skill require to set Pharmacy store & its management

E. Recommended Study Material

❖ **Text / Reference Books:**

1. Mohd. Aquil, Practice of Hospital, clinical & Community Pharmacy Elsevier
2. Paul Rutter, Community Pharmacy E-Book, Symptoms, Diagnosis and Treatment, 3rd Edition
3. A Textbook of Clinical Pharmacy Practice: G. Parthasarathi, Karin Nyfort-Hansen and Milap Nahata, Universities Press.
4. Research articles as per the assignment

ME781.01: OCCUPATIONAL HEALTH & SAFETY

Total Hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to occupational health and safety	02
2	Occupational safety and management standards, and regulations for health, safety and environment	04
3	Identifying safety and health hazards, and risk analysis	04
4	Occupational physiology and psychosocial factors, and work organization	04
5	Ergonomic workplace design and musculoskeletal diseases	04
6	Control of workplace hazards	04
7	E-waste management	04
8	Work practices in industries and global strategy on occupational safety and health	04

B. Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1	**Introduction to occupational health and safety**	**02 Hours**	**9%**
	Definition and history of occupational health & safety, workplace hazards.		
2	**Occupational safety & management standards and regulations for health, safety and environment**	**04 Hours**	**13%**
2.1	Factories act and rules; Workmen compensation act. Indian explosive act -		

	Gas cylinder rules - SMPV Act - Indian petroleum act and rules.		
2.2	Environmental pollution act. Manufacturing, storage and import of Hazardous Chemical rules 1989, Indian Electricity act and rules. Overview of OHSAS 18000 and ISO 14000 National legislation and public organizations.		
3	**Identifying safety and health hazards, and risk analysis**	**04 Hours**	**13%**
3.1	Hazard, risk issues and hazard assessment, Introduction to hazard, hazard monitoring-risk issue.		
3.2	Hazard assessment, procedure, methodology; safety audit, checklist analysis, what-if analysis, safety review, preliminary hazard analysis (PHA), hazard operability studies (HAZOP).		
3.3	Computer aided risk analysis, Fault tree analysis & Event tree analysis, Logic symbols, methodology, minimal cut set ranking - fire explosion and toxicity index (FETI), various indices – Hazard analysis (HAZAN).		
3.4	Failure Mode and Effect Analysis (FMEA), Basic concepts of software on risk analysis, CISCON, FETI, ALOHA.		
4	**Occupational physiology, Psychosocial factors and Work organization**	**04 Hours**	**13%**
4.1	Man as system component – allocation of functions – efficiency.		
4.2	Occupational work capacity aerobic and anaerobic work – evaluation of physiological requirements of jobs – parameters of measurements – categorization of job heaviness		
4.3	Work organization – stress – strain – fatigue – rest pauses – shift work – personal hygiene.		
5	**Ergonomic workplace design and Musculoskeletal diseases**	**04 Hours**	**13%**
5.1	Meaning of Ergonomic		
5.2	Meaning of Workplace Design.		
5.3	Musculoskeletal Diseases causes and prevention		
6	**Control of workplace hazards**	**04 Hours**	**13%**
6.1	Workplace hazards and risk control, Transport hazards and risk control, Musculoskeletal hazards and risk control, Work equipment hazards and risk		

	control		
6.2	Electrical safety, Fire safety, Chemical and biological health hazards and risk control, Physical and psychological health hazards and risk control, Health and safety practical application		
7	**E-waste management**	**04 Hours**	**13%**
7.1	Waste characteristics, generation, collection, transport and disposal		
8	**Work practices in industries and global strategy on occupational safety and health**	**04 Hours**	**14%**
8.1	Work practices in industries in manufacturing industries		
8.2	Work practices in industries in service industries		

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- Tutorials related to course content will be given to students.
- In the lectures discipline and behavior will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

D. Students Learning Outcomes:

- Students will be able to make models for safety at work.
- Students will be able to select safety methods.
- Students will be able to understand how hazardous the process is at work.
- Students will be able to understand proneness of accidents.

E. Recommended Study Material:

❖ **Text Books:**

1. Grimaldi and Simonds, Safety Management, AITBS Publishers, New Delhi (2001)

2. Industrial safety and health, David L. Goetsch, Macmillan Publishing Company, 1993.
3. R.K.Jain and Sunil S.Rao , Industrial Safety, Health and Environment Management Systems, Khanna publishers , New Delhi (2006)
4. Salvendy, G. (2012). Handbook of human factors and ergonomics. John Wiley & Sons.

❖ **Reference Books:**

1. Arezes, P., Baptista, J. S., Barroso, M. P., Carneiro, P., Cordeiro, P., Costa, N., & Perestrelo, G. (Eds.). (2013). Occupational Safety and Hygiene. CRC Press.
2. Chaturvedi, P. (2005). Managing Safety Challanges Ahead. Concept Publishing Company.
3. Healey, B. J., & Walker, K. T. (2009). Introduction to occupational health in public health practice (Vol. 13). John Wiley & Sons.
4. Hester, R. E., & Harrison, R. M. (2009). Electronic waste management (Vol. 27). Royal Society of Chemistry.
5. Karwowski, W., Soares, M. M., & Stanton, N. A. (Eds.). (2011). Human Factors and Ergonomics in Consumer Product Design: Uses and Applications. CRC Press.
6. Khan, B. H. (Ed.). (1997). Web-based instruction. Educational Technology.
7. Roughton, J., & Crutchfield, N. (2011). Job hazard analysis: A guide for voluntary compliance and beyond. Butterworth-Heinemann.
8. Salvendy, G. (Ed.). (2001). Handbook of industrial engineering: technology and operations management. John Wiley & Sons.
9. Smedley, J., Dick, F., & Sadhra, S. (Eds.). (2013). Oxford handbook of occupational health. Oxford University Press.
10. Tillman, C. (2006). Principles of occupational health and hygiene: an introduction. Allen & Unwin.

❖ **Web Material:**

1. International Labour Organization (ILO) http://www.ilo.org/public/english/
2. Occupational Safety & Health Administration United States Department of Laborhttps://www.osha.gov/about.html

- **Other Material:**

1. International Journal of Labour Research http://www.ilo.org/actrav/info/pubs/WCMS_158769/lang--en/index.htm
2. International journal of occupational safety And ergonomics (http://archiwum .ciop.pl/757).
3. Journal of Safety and Health at Work (http://www.journals.elsevier.com/safety-and-health-at-work/)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICAL SCIENCE

PD261 : Astrophysics, Space and Cosmos-1 (ASC–1)

University Elective
Semester 3 Undergraduate Programme
Semester 1 Postgraduate Programme

Prerequisite:

Student having exposure of 10+2 level of Physics, and studying in bachelor and master program from any institute can join this course.

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Basic Astronomical Techniques	10
2.	The Frontier of Space	06
3.	The Life and Death of Stars	08
4.	The Universe	06

Total Hours (Theory): 0

Total Hours (Lab): 30

Total Hours: 30

B. Detailed Syllabus:

1.	**Basic Astronomical Techniques**	**10 Hrs**
	Description of Telescopes, Methods of observation, electromagnetic spectral window, resolution, sensitivity, noise, signal to noise ratio, background, aberrations, Telescopes at different wavelengths, Detectors at different wavelengths, Imaging techniques, spectroscopy, calibration, Atmospheric effects at different wavelengths, Astronomical data analysis, H-R diagram, sun and solar system, Sky Gazing.	6 (L) + 4 (P)
2.	**The Frontier of Space**	**6 Hrs**
	Outer reaches of earth's Atmosphere, Earth's Atmospheric Layers and Orbital Mechanism: Types of Orbits Launching of Satellites, Basics of Satellite subsystems, channel and link. Satellite Applications: Earth Observation, Scientific Study, Weather Forecast, Military Applications, GPS.	
3.	**The Life and Death of Stars**	**8 Hrs**
	Stars and Galaxies, The story of collapsing stars: Star formation and evolution, Interstellar Nebula, Red Giant, Planetary Nebulae, Planetary Dynamics, White Dwarfs, Red super Giant, Supernovae- Neutron star, Black hole, naked singularity. Gravitational Lensing, Accretion disks Around Compact Objects, Binary pulsar, Gravitational Waves.	
4.	**The Universe**	**6 Hrs**
	Universe in different scale: solar scale, Galactic scale, Cosmological scale, vastness of our universe, Expanding universe (Big bang, inflation), some interesting facts of our universe.	

C. Instructional Methods and Pedagogy

The topics will be discussed in interactive class room sessions using classical black-board teaching, ICT, hands-on-experiments and demonstration of experiments, whichever is relevant. Assignments, small projects and lab-exercise will be to the students. Student’s needs to submit solution/report/results of above mentioned work, which will be eventually used for the evaluation purpose. Occasionally seminar/viva presentation will also be used as one of the evaluation tools.

D. Student Learning Outcomes:

Students will get preliminary knowledge about Astrophysics, Space and Cosmology. This is a level one course, which provides basis for the second level course. After completion of this course, students can start taking small project in this or allied fields.

F. Recommended Study Materials

- **Reference Book & Text Book:**

1. The Story of Collapsing Stars: Black Holes, Naked Singularities, and the Cosmic Play of Quantum Gravity, Prof. Pankaj S. Joshi
2. An Introduction to Cosmology 3ed, J.V. Narlikar
3. An introduction to modern cosmology, Andrew R. Liddle
4. Astronomy: Astronomy for Beginners: The Magical Science of Stars, Galaxies, Planets, Black Holes, Wormholes and much, much more! (Astronomy, Astronomy Textbook, Astronomy for Beginners), Miles Clarke
5. Satellite Communication, Dennis Roddy, Mc-Graw Hill publication
6. Satellite Communication, Timothy Pratt, Charles Bostian, Jeremy Allnut, John Wiley and Sons publication

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

List of University Elective Courses
For Second Semester Postgraduate Programme
(Academic year 2019-20)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	EE782.01: Energy Auditing and Management	EE/CSPIT
2	CE771.01: Project Management	CE/CSPIT
3	IT771.01: Cyber Security & Laws	IT/CSPIT
4	CA842:Mobile Application Development	CMPICA
5	PT796.01: Fitness and Nutrition	ARIP
6	NR 752.01: Epidemiology & Community Heath	MTIN
7	OC733.01: Introduction to Polymer Science	PDPIAS
8	MB651: Software based Statistical Analysis	I2IM
9	MA772.01: Design of Experiments	PDPIAS
10	PH892: Intellectual Property Rights	RPCP
11	PD262: Astrophysics, Space and Cosmos-2 (ASC-2)	PDPIAS

EE782.01: ENERGY AUDITING & MANAGEMENT

Total hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the Course:

Sr. No.	Title of Unit	Min. No. of Hrs
1	Electrical Energy Conservation	3
2	Electrical Energy Management	10
3	Financial Management	4
4	Energy Management & Audit	5
5	Energy Efficient Technologies In Electrical Systems	3
6	Energy Storage Systems	3
7	Case Studies.	2

B. Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1	**Electrical Energy Conservation**	**3 Hours**	**11.33%**
1.1	**Energy Scenario:** Introduction to energy science and energy technology, various forms of energy. Law of conservation of energy. Energy scenario of India, Introduction to global energy scenario. Carbon credit, Energy Sector Reforms, Energy Strategy for the Future, Energy Conservation Act 2001 and its features.		
1.2	**Measures for energy conservation:** Potential energy conservation opportunities in: HVAC System, Lighting systems, Motors and Transformers.		
2	**Electrical Energy Management**	**10 Hours**	**29.12%**
2.1	Concept of energy management, Design of Energy management programmes, energy cost, Energy planning, Energy staffing, Energy Organization, Energy Requirement, Energy		

	Costing, Energy Budgeting, Energy Monitoring, Energy consciousness, Energy Management Professionals, Environment pollution due to energy use. Need of energy planning, steps for energy planning, Energy management in industry, Energy management cell function and objective, Energy management cell roles and responsibilities, Role of energy manager, benchmarking, Social and economic cost benefits. Seven principals of energy management.		
2.2	**Electrical System Optimization** Electricity rate tariff, key to reduction in electrical energy Consumption, Methods to improve plant power factor, load management, conduction loss, switching loss, magnetic loss, harmonic Compensation, Motor control, Lighting energy saving.		
2.3	**Cogeneration** Definition, Need, Application, Advantages, Classification, Saving potentials		
3	**Financial Management**	**4 Hours**	**11.11%**
	Investment-need, Appraisal and criteria, Financial analysis techniques-Simple payback period, Return on investment, Net present value, Internal rate of return, Cash flows, Risk and sensitivity analysis; Role of ESCOs.		
4.	**Energy Management & Audit**	**5 Hours**	**12.89%**
	Introduction, Definition, Energy audit- needs, types and walkthrough energy audit. Energy audit at unit level, Industrial Audit approaches. Procedure for energy audit and equipments required. Comprehensive Energy audit Site testing Measurement & Analysis of Electrical System like Induction Motors. Transformers, Illumination system, Problems on Energy Management.		
5.	**Energy Efficient Technologies In Electrical Systems**	**3 Hours**	**11.11%**
	Load Management and Maximum demand control. Electrical distribution system. Maximum demand controllers, Automatic power factor controllers, Energy efficient motors, Soft starters with energy saver, Variable speed drives, Energy efficient transformers, Electronic ballast, Occupancy sensors, Energy efficient lighting controls.		
6.	**Energy Storage Systems**	**3 Hours**	**13.33%**
	Introduction, Demand for energy storage, Energy storage systems: heat storage- hot water, hot solids, phase change materials; Potential energy storage: spring, compressed gas. Pumped hydro: Flywheels. Rolling mills, Electrical and magnetic energy storage systems.		

7.	**Case Studies.**	**2 Hours**	**11.11%**

C. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a component of the overall evaluation.
- Minimum two internal exams will be conducted and will be considered as a part of overall evaluation.
- Assignments based on course content will be given to the students for each unit/topic and will be evaluated at regular interval and its weightage may be reflected in the overall evaluation.

D. Student Learning Outcomes:

After the completion of the course the students will be able to:

1. Analyze about energy scenario nationwide and worldwide
2. Decide about energy management in more effective way.
3. To propose the effective way for energy conservation
4. Carry out financial management.
5. Can design energy efficient technologies and provide alternative solutions for energy storage

E. Recommended Study Material:

❖ **Text Book:**

1. Amlan Chakrabarti, Energy engineering and management, PHI Learning Private Limited.
2. K. Nagabhusan Raju, Industrial Energy Conservation Techniques, Atlantic Publishers & Distributors (P) Ltd.

❖ **Reference Book:**

1. Renewable energy sources and conservation technology By- N.K.Bansal, Kleemann and Meliss
2. Non – conventional energy sources by G.D. Rai
3. Energy technology by S.Rao.
4. A guide to energy management by Barney L Capehart, William J Kennedy, Wayne C Turner.

❖ **Web Material:**

1. www.energymanagertraining.com
2. www.bee-india.gov.in

CE771.01: PROJECT MANAGEMENT

Total hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	**-**	2	2	2
Marks	-	100	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Overview of Project Management	03
2	Project Management Concepts and Techniques	03
3	Project Cost Estimation	04
4	Project Planning and Scheduling	05
5	Project Monitoring and Control	05
6	Material Management in Project	04
7	Management of Special Projects and Project Management Software Tools	06

Total hours (Theory): 30

B. Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Overview of Project Management**	**03 Hours**	**10 %**
	Introduction to Project Management Overview of Project Planning, Project Estimation, Project Scheduling, Organization and Team Structure, Risk Analysis and Management, Resource Allocation Project Management Process and Role of Project Manager		
2.	**Project Management Concepts and Techniques**	**03 Hours**	**10 %**
	Project Screening and Selection Techniques		

	Structuring Concepts and Tools (WBS,ORS,LRC) Project Planning Tools (Bar Chart, LOB, CPM and PERT) Risk Analysis and Management		
3.	**Project Cost Estimation**	**04 Hours**	**13 %**
	Types of Estimates and Estimating Methods Project Budgeting		
4.	**Project Planning and Scheduling**	**05 Hours**	**17 %**
	Dynamic Project Planning and Scheduling Project Scheduling with Resource Constraints		
5.	**Project Monitoring and Control**	**05 Hours**	**17 %**
	Monitoring Techniques and Time control system Project Cost Control and Time cost tradeoff		
6.	**Material Management in Project**	**04 Hours**	**13 %**
	Project Procurement and Material Management		
7.	**Management of Special Projects and Project Management Software Tools**	**06 Hours**	**20 %**
	Management of SE/NPD/R&D/Hi-Tech and Mega Projects Software tools for Project Management: MS Project, Primavera, Turbo Project, Riski Project.		

C. Instructional Method and Pedagogy:

- Lectures will be taken in class room with the use of multi-media presentations and black board – mix of both.
- Assignments based on above course content will be given to the students at the end of each chapter. Each assignment contains minimum 5 questions.
- Quizzes and Surprise tests will be conducted for testing the knowledge of students for particular topic.

D. Student Learning Outcomes:

Upon successful completion of this course, students will be able to understand project management process and different aspect of development process necessary for the management of the project which includes various activities, resources, quality, cost and system configuration etc.

E. Recommended Study Material:

❖ **Text Books:**

1. Project management: engineering, technology, and implementation by Shtub, Avraham, Jonathan F. Bard, and Shlomo Globerson, Prentice-Hall, Inc., 1994.
2. Project Management Handbook by Lock, Gower.
3. VNR Project Management Handbook by Cleland and King.
4. Management guide to PERT/CPM by Wiest and Levy, PHI.
5. Project Management: A Systemic Approach to Planning, Scheduling and Controlling by Horald Kerzner, CBS Publishers, 2002.
6. Project Scheduling and Monitoring in Practice by S. Choudhury,
7. Total Project Management: The Indian Context by P. K. Joy, Macmillan India Ltd.

❖ **Reference Books:**

1. Project Management for Business and Technology: Principles and Practice by John M Nicholas, Prentice Hall of India, 2002.
2. Project Management, by N. J. Smith (Ed), Blackwell Publishing, 2002.
3. Effective Project Management by Robert K. Wysocki, Robert Back Jr. and David B. Crane, John Wiley, 2002.
4. Project Management: A Managerial Approach, by Jack R Meredith and Samuel J Mantel, John Wiley, 4th Edition, 2000.

IT771.01: CYBER SECURITY & LAWS

Total hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	02	02	02
Marks	-	100	100	

A. Outline of the Course:

Sr No.	Title of the unit	Minimum number of hours
1.	Computer and Cyber Security Basics	06
2.	Security Threats	09
3.	Provisions in Indian Laws in dealing with Cyber Crimes	07
4.	Case Studies	08

Total hours (Theory): 30

B. Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Computer and Cyber Security Basics**	**06 hours**	**20 %**
	Introduction to Computers, Computer History, Software, Hardware, Classification, Computer Input-Output Devices, Windows, DOS Prompt Commands, Basic Computer Terminology, Internet, Networking, Computer Storage, Computer Ethics and Application Programs, Security : Security trends –Goal, Attacks, Services and Mechanism		
2.	**Security Threats**	**09 hours**	**30 %**
	Application security (Database, E-mail and Internet), Data Security Considerations-Backups, Archival Storage and Disposal of Data, Security Technology-Firewall and VPNs, Intrusion Detection, Access Control. Security Threats -Viruses, Worms, Trojan Horse, Bombs, Trapdoors, Spoofs, E-mail viruses, Macro viruses, Malicious Software, Network and Denial of		

	Services Attack, Security Threats to E-Commerce- Electronic Payment System, e-Cash, Credit/Debit Cards. Digital Signature, public Key Cryptography.		
3.	**Provisions in Indian Laws in dealing with Cyber Crimes**	**07 hours**	**23 %**
	Security Policies, Why Policies should be developed, WWW policies, Email Security policies, Policy Review Process-Corporate policies-Sample Security Policies, Publishing and Notification Requirement of the Policies. Information Security Standards-ISO, IT Act, Copyright Act, Patent Law, Cyber Laws in India; IT Act 2000 Provisions, Intellectual Property Law.		
4.	**Case Studies**	**08 hours**	**27 %**
	Identity Management, Cyber Security and Terrorism: Case Studies, The DigiNotar case, Deutsche Telekom, The disruption at the IT service provider Tieto, Web Based Attacks by Symantec, Password Security		

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- **Assignments** based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of **5 Marks** as a part of internal theory evaluation.
- Surprise Tests/Quizzes/Seminar/**Case Study** will be conducted which carries **10 Marks** as a part of internal theory evaluation.

D. Student Learning Outcome:

Learning outcomes of the course are:

- Students will able to do classification of cybercrime, methods used to perform crime, apply cyber security, and know the detailing of Information Technology Acts against offences.

- Students will understand and appreciate the legal and ethical environment impacting individuals as well as business organizations and have an understanding of the ethical implications of IT legal decisions.
- Students will have a fundamental knowledge of Information Technologies which affect organizational processes and decision-making.

E. Recommended Study Material:

❖ **Reference Books:**

1. Charles P. Pfleeger, Shari Lawerance Pfleeger, "Analysing Computer Security", Pearson Education India.
2. V.K. Pachghare, "Cryptography and information Security", PHI Learning Private Limited, Delhi India.
3. Dr. Surya Prakash Tripathi, Ritendra Goyal, Praveen kumar Shukla,"Introduction to Information Security and Cyber Law" Willey Dreamtech Press.
4. Schou, Shoemaker, " Information Assurance for the Enterprise", Tata McGraw Hill.
5. Chander, Harish, "Cyber Laws And It Protection" , PHI Learning Private Limited ,Delhi, India

CA 842: MOBILE APPLICATION DEVELOPMENT

Total hours (Theory): 30

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the Course:

Week No	Practical	Description
1	Brief about mobile technologies and challenges in mobile application development and architectural overview of an Android platform.	Different mobile application development platform overview and Android architecture overview and basic components of Android application development overview.
2	Development of first Android based mobile application and overview of necessary components required for development.	Working of Android Studio IDE, Android project directory structure, Dalvik Virtual Machine Overview, Android Software development kit explanation, Virtual device creation and execution of first application on virtual as well as actual device
3	Fundamentals of User Interface designing.	XML based user interface designing using different available layouts like Relative Layout, Linear Layout, Table Layout etc.
4	User Interface Widgets-1	Hands-on demonstration of basic widgets like- TextView, EditText, Button, ToggleButton, RadioButton, RadioGroup, CheckBox, RatingBar, SeekBar etc.
5	User Interface Widgets-2	Hands-on demonstration of composite widgets like- ListView, Spinner and

		AutoCompleteTextView and customization of the composite controls.
6	Activity, Activity navigation and Intents.	Activity life cycle, Linking Activity using Intents:startActivity(), StartActivityForResult(). Calling built-in applications: ACTION_MAIN, ACTION_VIEW, ACTION_DIAL, ACTION_SEND
7	Android Resources, Styles and Themes.	Usage and implementation of different resources like drawable, string, color, dimes, raw and animation. Creating and Applying simple Style, Inheriting built-inStyle and User defined style, Using Styles as themes.
8	Dialogs & Menus.	Hands-on demonstration of different dialogs and menus available in Android.
9	Data Persistence Techniques.	User Preferences and Database management through SQLite
10	Broadcast Actions and Services implementation.	Service: life cycle, create and destroy service, Alarm Manager and SMS Manager. Standard Broadcast Actions.
11	PHP based web service implementation in Android	Creation and consumption of PHP based web service.
12	Simple Google Map incorporation with Android application.	Google Developer console usage, SHA-1 certificate creation and API-KEY creation and incorporation in Android application.

❖ **Text Books:**

1. Wei-Meng Lee: Beginning Android 4 Application Development, Wiley India Pvt Ltd.
2. Mark L. Murphy: The Busy Coder's Guide to Android Development

Reference Books:

1. Jonathan Simon: Head First Android Development, O'REILLY publication
2. Mark L Murphy: Beginning Android, Wiley India Pvt. Ltd.

Web References:

1. https://developer.android.com [Detail Android Development Guide]
2. https://www.youtube.com/watch?v=SUOWNXGRc6g&list=PL2F07DBCDCC01493A [200 android development tutorials]
3. www.androidhive.info/ [Advance application development with Android]

PT796.01: FITNESS AND NUTRITION

Total hours (Theory): 30

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Physical Education and Physical Fitness	10
2.	Nutrition and Health	10
3.	Sports and Life Skills Education	10

B. Detailed syllabus:

Sr. No.	UNIT	Hours
1	**PHYSICAL EDUCATION AND PHYSICAL FITNESS**	**10 hours**
1.1.	Concept of Physical Education, Meaning, Definition, Aims and Objectives of Physical Education, Need and Importance of Physical Education	
1.2	Physical Education and its Relevance in Inter Disciplinary Context	
1.3	Physical Fitness Components: Type of Fitness, Health Related Physical Fitness, Performance Related Physical Fitness	
2	**NUTRITION AND HEALTH**	**10 hours**
2.1	Concept of Food and Nutrition, Balanced Diet, Food Pyramid Index	
2.2	Macro and Micronutrients: Types, functions and classification system	
2.3	Carbohydrates: Types, RDA data, Glycemic index, Sources of Fats, saturated, unsaturated fats, recommended intake, importance of fat in diet, fats in health and disease	

2.4	Protein: Types EAA, function, assessing quality of proteins, selecting incomplete proteins, RDA sources.	
2.5	Vitamins And Minerals: Types, functions, sources, and minerals - calcium, Phosphorus , iron, magnesium, sodium, potassium, and chloride. Trace elements - sources and functions.	
2.6	Determining Caloric Intake and Expenditure, Obesity, Causes and Preventing Measures – Role of Diet and Exercise, Importance of hydration in exercise	
3	**SPORTS AND LIFE SKILLS EDUCATION**	**10 hours**
3.1	Sports and Socialization	
3.2	Physical Activity and Sport – Emotional Adjustment and Wellbeing	
3.3	Substance Abuse among Youth – Preventive Measures and Remediation	
3.4	Yoga, Meditation and Relaxation	
3.5	Sports and Character Building	
3.6	Values in Sports	
3.7	Sports for World Peace and International Understanding	

C. Instructional Method And Pedagogy:

- Interactive classroom sessions using black-board and audio-visual aids.
- Using the available technology and resources for e-learning.
- Students will be encouraged towards self-learning and under direct interaction with course faculty.
- Students will be enabled for continuous evaluation.
- Case study, didactic mode of group discussions

D. Student Learning Outcomes:

Upon completion of the course, the student should be able to :

- Appraise the importance of exercise in maintenance of health and fitness.
- Objectively define health and fitness in realistic environment and inculcate habits of physical activity, nutrition and sports as a behavior change and overall health promotion

E. Recommended Study Material:

❖ **Textbooks:**

1. ACSM"s "Health Related Physical Fitness Assessment Manual Lippincott Williams and Walkins USA, 2005.
2. Siedentop.D,(1994) Introduction to Physical Education and Sports (2nd ed.), California: Mayfield Publishing Company.
3. Corbin.Charles Beetal. C.A., (2004) Concepts of Fitness and Welfare Boston McGraw Hill.

NR 752.01: EPIDEMIOLOGY & COMMUNITY HEALTH

Total hours (Theory): 30

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the Course:

Unit No.	Title of Unit	Prescribed Hours
1	**Introduction:** • Concept, scope, definition, trends, History and development of modern Epidemiology • Contribution of epidemiology • Implications	5
2	**Health Statistics:** • Morbidity & Mortality	2
3	**Epidemiological approaches:** • Study of disease causatives (Cause & Risk) • Health promotion • Levels of prevention	4
4	**Epidemiology of** • Communicable diseases • Non-communicable diseases	10
5	**Disaster:** • Disaster preparedness, • Disaster management	3
6	**Health Organizations:** • Voluntary health organizations • International health agencies –WHO, World health assembly, UNICEF, UNFPA, SIDA, US AID,	6

	DANIDA, DFID. AusAID etc	
	Total	**30 Hours**

B. Instruction Method and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, microteaching, task-based learning, assignments, field visit and various interpersonal activities like group work, independent and collaborative research, presentations etc.

C. Evaluation:

The students will be evaluated continuously in the form of internal as well as external examinations. The evaluation (Theory) is schemed as 25 marks for internal evaluation and 75 marks for external evaluation in the form of University examination.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sr. No.	Component	Number	Marks per incidence	Total Marks
1	Assignments	1	8	8
2	Internal Test/ Model Exam	1	12	12
3	Attendance and Class Participation	Minimum 80% attendance		10
			Total	**30**

External Evaluation

The University examination will be for 70 marks and will be based on practical based tests and a viva-voce.

D. Learning Outcomes:

At the end of the course, learners will be able to:

- Understand the epidemiologic terminology,

- Understand the health statistic
- Understand the various methods of epidemiology
- Understand the role and functions of various health agencies
- Understand the epidemiological trends in communicable and non-communicable diseases.

OC733.01: INTRODUCTION TO POLYMER SCIENCE

Total hours (Theory): 30

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the course

Sr.No	Title of Unit	Approximate No of Hours
1	Basic concepts of Polymer Chemistry	5
2	Chemistry of Polymerization	8
3	Kinetics of polymerization	8
4	Molecular weight and size	9

B. Detailed syllabus:

Sr. No	Title of Unit	Approximate No of Hours
1.	**Basic concepts of Polymer Chemistry**	**5**
	Introduction to polymers, How are polymers made, Classification of polymers.	
2.	**Chemistry of Polymerization**	**8**
	Chain polymerization, Step polymerization, Miscellaneous polymerization reactions, Polymerization techniques.	
3.	**Kinetics of polymerization**	**8**
	i). Free – Radical chain polymerization. ii). Cationic & anionic polymerization. iii). Polycondensation	
4.	**Molecular weight and size**	**9**
	Number – Average molecular weight. Viscosity – Average molecular weight. Polydispersity and molecular weight.	

	Significance of polymer molecular weight. Size of polymer molecules	

C. Instrumental Methods and Pedagogy:

Topics will be taught in interactive class room sessions using black-board and if required power-point presentations will also be employed. Special interactive problem solving sessions will be conducted. Course materials will be provided from various sources of information. Students will be trained to measure molecular weight of polymers using appropriate instrument(s). Unit test will be taken regularly as a part of continuous evaluation and suggestions will be given to the students in order to do better in their performance.

D. Student Learning Outcomes/Objective:

- The programme aims at providing the basic concepts in polymer science.
- Ensuring that students acquire skills for further research in this area.

E. References:

1. A First Course in Polymer Chemistry by A. Strepikheyev , V.Derevitskaya and G.Slonimsky ; MIR Publishers, Moscow
2. Polymer Science by V.R.Gowariker , N.V.Viswanathan and Jayadev Sreedhar, New Age International Publishers.
3. Polymer Science and Technology of Plastics and Rubbers by Premamoy Ghosh, Tata McGraw-Hill Publishing Company Ltd. New Delhi.

MB651: SOFTWARE BASED STATISTICAL ANALYSIS

Total hours (Theory): 30

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**Introduction to Statistics** • Research and Innovation • Introduction to Statistics • Quantitative Techniques in Research	**06**
2	**Software # 1 - Use of Software in Research** • Introduction to the Software • Creating Variables • Data and its Types • Importing Data from MS-Excel • Transformation of Variables • Visual Benning • Determining Validity and Reliability of Scale using CFA - Cronbach's (alpha) : a coefficient of internal consistency • Estimation & Hypothesis Testing: • Parametric Tests ○ Z Test ○ t – Test ○ Cross Tabulation and Chi – square	**12**

<table>
<tr><th>Module No.</th><th>Title/Topic</th><th>Classroom Contact Sessions</th></tr>
<tr><td></td><td><li style="list-style:none">
One-Way & Two-Way Analysis of Variance (ANOVA)
Pearson’s Correlation Analysis
Regression Analysis
Simple & Multiple Linear Regression Analysis
Measures of Model Fit (R and R-square Statistics)

Non-Parametric Tests
Mann-Whitney U Test
Wilcoxon Signed Rank Test
Run test
Krushal-Wallis Test
Spearman Correlation Analysis
</td><td></td></tr>
<tr><td>3</td><td>Statistical Analysis Using Open Source software
Introduction to Software
Programming Language Basics – including creating, sub-setting and analyzing
Managing your files and workspace
Controlling functions (procedures or commands)
Data Acquisition – Reading files
Data Transformations
Selecting variables and observations
Writing functions (macros)
Graphics
</td><td>10</td></tr>
<tr><td>4</td><td>Article / Research Papers Reviews</td><td>02</td></tr>
<tr><td colspan="2">Total</td><td>30</td></tr>
</table>

B. Pedagogy

The course will emphasise self-learning and active classroom interaction based on students’ prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session.

Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will be as follows:

- Classroom / Practical Contact Sessions … About 28 Sessions
- Research Paper Discussions / Feedback … About 02 Sessions

The exact division among the above components will be announced by the instructor at the beginning of the semester as a part of detailed session-wise schedule.

C. Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks	Percentage of total internal evaluation
1	Quizzes	3	10	30	10
2	Assignment / Project work	1	150	150	50
3	Internal Tests	2	45	90	30
4	Attendance and Class Participation			30	10
			Total	**300**	**100**

The total marks will be divided by 10 and declared as Institute-level evaluation marks for the course. The Institute-level evaluation will constitute 30% of the total marks f or the course.

D. External Evaluation

The University examination will be for 70 marks and will be based on practical computer-based tests and a viva-voce.

E. Learning Outcomes

At the end of the course, the student should have developed:

- Skills related to use of statistical techniques for analysis using software

- Rational decision making skills for typical business / other decisions
- Inputs for reviewing articles / research papers especially related to use of statistical techniques and analysis based on software

F. Reference Material

- ❖ **Text-book**

1. Latest Manuals of Software

- ❖ **Reference Books**

1. David .M. Levine, Krehbiel, Berenson, P.K. Viswanathan, (Latest Edition), Business Statistics – A First Course, (Latest Edition), Pearson Education

MA 772.01: DESIGN OF EXPERIMENTS

Total hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	**-**	2	2	2
Marks	-	100	100	

A. Outline of the course:

Sr No.	Title of the unit	Minimum number of hours
1.	Principles of Experimental Design	06
2.	Statistical Concepts	06
3.	Single Factor Experiments	08
4.	Factorial Experiments	10

B. Detailed Syllabus:

Sr. No.	UNIT	Hours
1	**Principles of Experimental Design 06 Hours**	**20%**
1.1	Basic Terminologies: Types of Investigations and Experiments	
1.2	Confirmatory and Exploratory Experiments	
1.3	Modeling and selecting Response	
1.4	Minimizing Bias and Variability	
2	**Statistical Concepts 06 Hours**	**20%**
2.1	Descriptive Statistics and Graphical Presentation	
2.2	Probability Distributions, Hypothesis Tests and Confidence Intervals	
2.3	Power and Sample size calculation	
2.4	Experiments for Two Treatments	
2.5	Linear Regression: Simple and Multiple	

3.	**Single Factor Experiments** **08 Hours**	**30%**
3.1	Completely Randomized Designs	
3.1	Concepts of Multiple comparison	
3.2	Pairwise Comparisons	
3.3	Comparisons with a Control	
3.4	General Contrast	
4.	**Factorial Experiments** **10Hours**	**30%**
4.1	Inference from Factorial Experiments	
4.2	Two-Level Factorial Experiments	
4.3	Definition and Estimation of Main Effects and Interactions	
4.4	Statistical Analysis	
4.5	Two-Level Fractional Factorial Experiments: Introduction	

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures/laboratory which carries a 5% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weighting of 5%.
- Two Quizzes (surprise test) will be conducted which carries 5% component of the overall evaluation.

D. Student Learning Outcomes:

- At the end of the course the students will be able to understand the basic concepts of Design of Experiment.
- Student will be able to apply concepts of these course in their study of specialization

E. Recommended Study Material:

❖ **Text Books:**

1. Tamhane, Ajit C. Statistical analysis of designed experiments: theory and applications. Vol. 609. John Wiley & Sons, 2009.
2. Hinkelmann, Klaus, and Oscar Kempthorne. Design and Analysis of Experiments, Introduction to Experimental Design. Vol. 1. John Wiley & Sons, 2008.
3. Lorenzen, Thomas, and Virgil Anderson, eds. *Design of experiments: a no-name approach*. CRC Press, 1993.

❖ **Reference Books:**

1. Cox, David Roxbee, and Nancy Reid. *The theory of the design of experiments*. CRC Press, 2000.
2. Goupy, Jacques L. *Methods for experimental design: principles and applications for physicists and chemists*. Vol. 12. Elsevier, 1993.

PH892: INTELLECTUAL PROPERTY RIGHTS

Total Hours (Theory): 30

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. **Outline of the course:**

Sr. No.	Title of Unit	No. of Contact Hours.
1	Intellectual Property Concepts	07
2	Practical aspect of patenting	20
3	IPR related treaties	03

B. **Syllabus Topics:**

Sr. No.	Title of Unit	Topics
1	Intellectual Property Concepts	• Concept of property, conventional property vs. Intellectual Property • Basic aspect of the 8 different IPR mechanism Viz. Patents, Copyright, trademark, industrial design, layout design of integrated circuits, geographical indicators, plant varieties & trade secrets. • Benefits of IPRs to improve the quality of research work • Strategies for avoiding research duplications, infringements
2	Practical aspect of patenting	• Indian patent act and its recent amendment with respect to following aspect: Patentable and non-patentable inventions, Essential criteria for filing a patent, Filling a patent in India and abroad, Drafting of patent application • Patent Filling and Commercialization: Procedure for patent obtaining in India, National and International Patent Search,

		Patent Analysis, Patent Drafting and Filling Procedure in India, Patent Specification and Claims, How to right a Claim of Patent, Pre/Post Grant Issues in Patenting, Opposition of Patent Granting, Infringement Analysis, Ground of Defense, Intellectual Property Appellation Board (IPAB), International Filing: PCT System • Patent and Biodiversity Act • Introduction to World Intellectual Property Organization. (WIPO) • Commercialization of patent: Need for Commercialization of research and role of IPRs in research Commercialization. • Benefit/Disadvantages of patenting to the society • Latest Amendment/Emerging Issues in Patenting • Case Study
3	IPR related treaties	• Patent co-operative treaty • Budapest treaty

C. Instructional Methods and Pedagogy:

The course employs in interactive classroom session using chalk and talk teaching to power point presentations. It also includes presentation by students on a specific topic assigned to them by the faculty and case study discussion with various litigation, infringement and patent rejection cases. Unit test will be conducted regularly as a part of continuous evaluation and suggestion will be given to student in order to improve their performance.

D. **Students Learning Outcomes/Objectives:**

- At the end of the course, the student will be able to understand the fundamental concepts of Intellectual Property Rights which further will be helpful in understanding other advanced aspects of Patent and Trademark applications in various scientific research and innovation.
- At the end student would gain experience in filling and drafting procedure of Patent.

E. **Recommended Study material**

1. Intellectual Property Right basic Concept, by M. M. S. Khatri, Atlantic Publisher and Distributors Pvt. Ltd., New Delhi.
2. Epstein on Intellectual Property: 5th Edition by Michael M. Epstein, Wolters Kulwer India Pvt. Ltd. Gurgaon, India.
3. Intellectual Property Right and Human Development in India by Shabana Talwar, First Edition, Serials Publications, New Delhi, India.
4. IPR Handbook for Pharma Students and Researchers, Parikshit Bansal, Pharma Book Syndicate, Hyderabad, India.
5. Patents, N. R. Subbaram, Pharma Book Syndicate, Hyderabad.
6. Intellectual Property Right by Nikolaus Thumm, Springer-Verlag Publications, Germany.
7. Intellectual Property - Patents, Copyright, TradeMarks and Allied Rights by Cornish, Aplin and Llewelyn, Sweet and Maxwell – Thomson Publishers, New Delhi, India
8. The Enforcement of Intellectual Property Rights: A Case Book by Louis Tc. Harms, WIPO Publishing House, Geneva.
9. Intellectual Property: From Creation to Commercialization - A Practical Guide for Innovators & Researchers by John P., MC Manus, Oak Tree Press, Ireland.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICAL SCIENCE

PD262 : Astrophysics, Space and Cosmos-2 (ASC-2)

University Elective

Semester 4 Undergraduate Programme
Semester 2 Postgraduate Programme

Prerequisite:

Student having exposure of 10+2 level of Physics, and studying in bachelor and master program from any institute can join this course.

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	**-**	**2**	**2**	**2**
Marks	**-**	**100**	**100**	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Black holes and Shadows	10
2.	Gravitational Waves	06
3.	Sensors and Detectors for Satellite Imaging	08
4.	Cosmology	06

Total Hours (Theory): 0

Total Hours (Lab): 30

Total Hours: 30

B. Detailed Syllabus:

1.	**Black holes and Shadows**	**10 Hrs**
	Spacetime, Schwarzschild metric, Null Geodesics in Schwarzschild metric, Photon sphere, Einstein ring, Shadow of Black hole, shadow of Naked singularity, recent observation of black hole image (M87 Galaxy), experiments using imaging technique	8 (L) + 2 (P)
2.	**Gravitational Waves**	**6 Hrs**
	Introduction to Gravitational Waves; recent development for the detection of Gravitational waves; Overview of Gravitational Wave detectors (i.e. LIGO, KAGRA, etc.) and its scientific and technologic importance; Outline of Interferometers & its applications; Construction and working of Michelson Interferometer & its application potentiality in Advance- LIGO; Demonstration experiment of Michelson Interferometer.	4 (L) + 2 (P)
3.	**Sensors and Detectors for Satellite Imaging**	**8 Hrs**
	Introduction to sensors, Signal, Parameters, Types of Sensors: CCD and CMOS, CMOS sensors: Architecture Configuration and Working, CCD sensors: Architecture Configuration and Working.	
4.	**Cosmology**	**6 Hrs**
	Galaxies: General Description, The Milky-Way, M87 Galaxy, Rotation Curves of Spiral Galaxies, Galaxy Formation, Expanding Universe, Hubble's Law, Dark matter and Dark Energy.	

C. Instructional Methods and Pedagogy

The topics will be discussed in interactive class room sessions using classical black-board teaching, ICT, hands-on-experiments and demonstration of experiments, whichever is relevant. Assignments, small projects and lab-exercise will be to the students. Students needs to submit solution/report/results of above mentioned work, which will be eventually used for the evaluation purpose. Occasionally seminar/viva presentation will also be used as one of the evaluation tools.

D. Student Learning Outcomes:

After completion of Astrophysics, Space and Cosmology level one and two courses, students will explore themselves for taking projects, which may eventually fulfill their curiosity in these and allied fields. It may open up new avenues in their career.

E. Recommended Study Materials

- **Reference Book & Text Book:**

1. The Story of Collapsing Stars: Black Holes, Naked Singularities, and the Cosmic Play of Quantum Gravity, Prof. Pankaj S. Joshi
2. An Introduction to Cosmology 3ed, J.V. Narlikar
3. An introduction to modern cosmology, Andrew R. Liddle
4. Astronomy: Astronomy for Beginners: The Magical Science of Stars, Galaxies, Planets, Black Holes, Wormholes and much, much more! (Astronomy, Astronomy Textbook, Astronomy for Beginners), Miles Clarke
5. Satellite Communication, Dennis Roddy, Mc-Graw Hill publication
6. Satellite Communication, Timothy Pratt, Charles Bostian, Jeremy Allnut, John Wiley and Sons publication.
7. CMOS/CCD Sensors and Camera Systems, Second Edition, Gerald Holst, Terrence S Lomheim, SPIE publication.
8. Review article: Advanced LIGO, The LIGO Scientific Collaboration, Classical and Quantum Gravity, Volume 32, Number 7 (2015)